statshw8

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Question 17

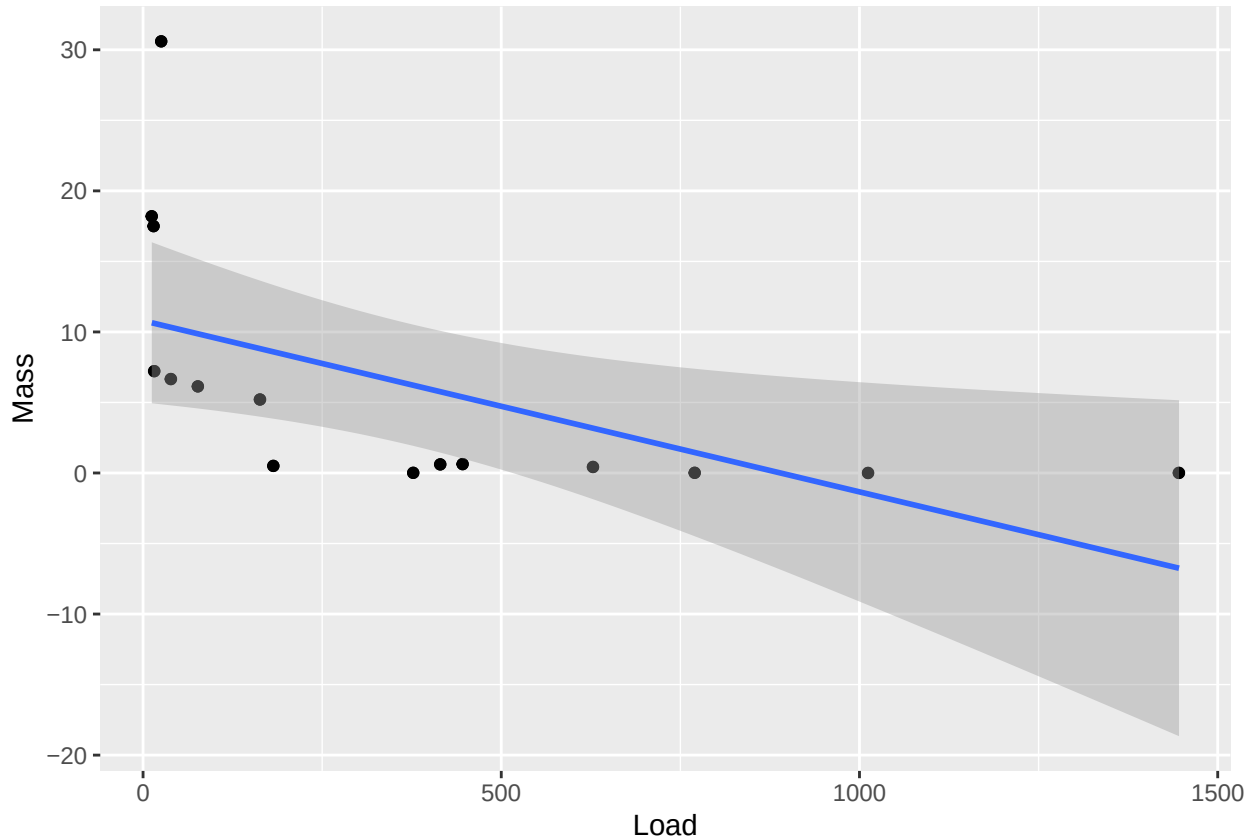
part a

```r
library(dplyr)
```

```
##
## Attaching package: 'dplyr'

## The following objects are masked from 'package:stats':
##
##     filter, lag

## The following objects are masked from 'package:base':
##
##     intersect, setdiff, setequal, union
```

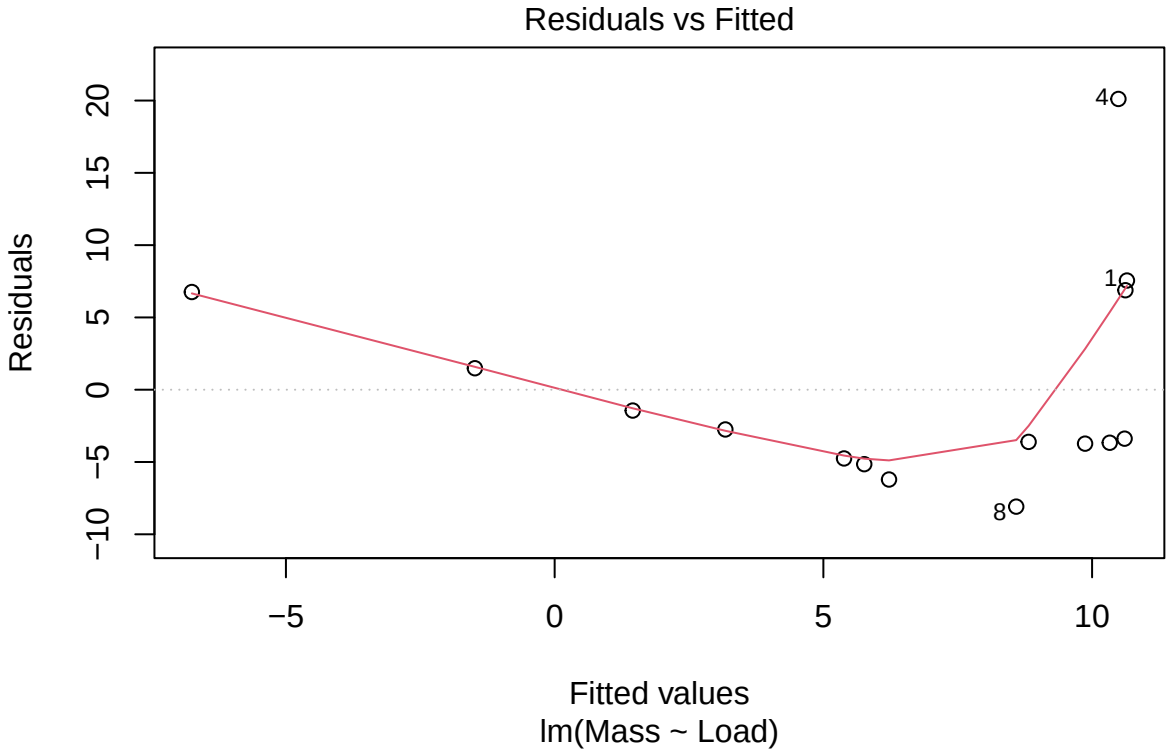
```r
library(Sleuth3)
library(ggplot2)

ex0817 |> ggplot(aes(x=Load, y= Mass)) + geom_point() + geom_jitter() +
  geom_smooth(method=lm)
```

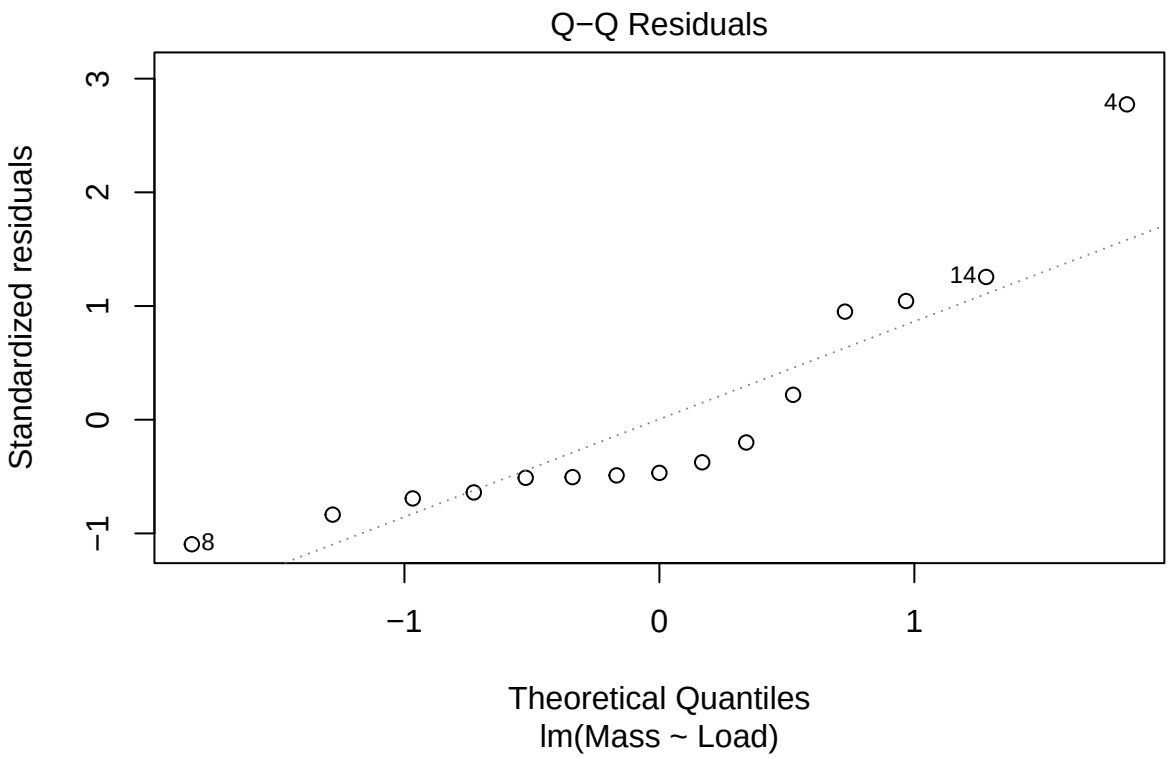
```
## `geom_smooth()` using formula = 'y ~ x'
```

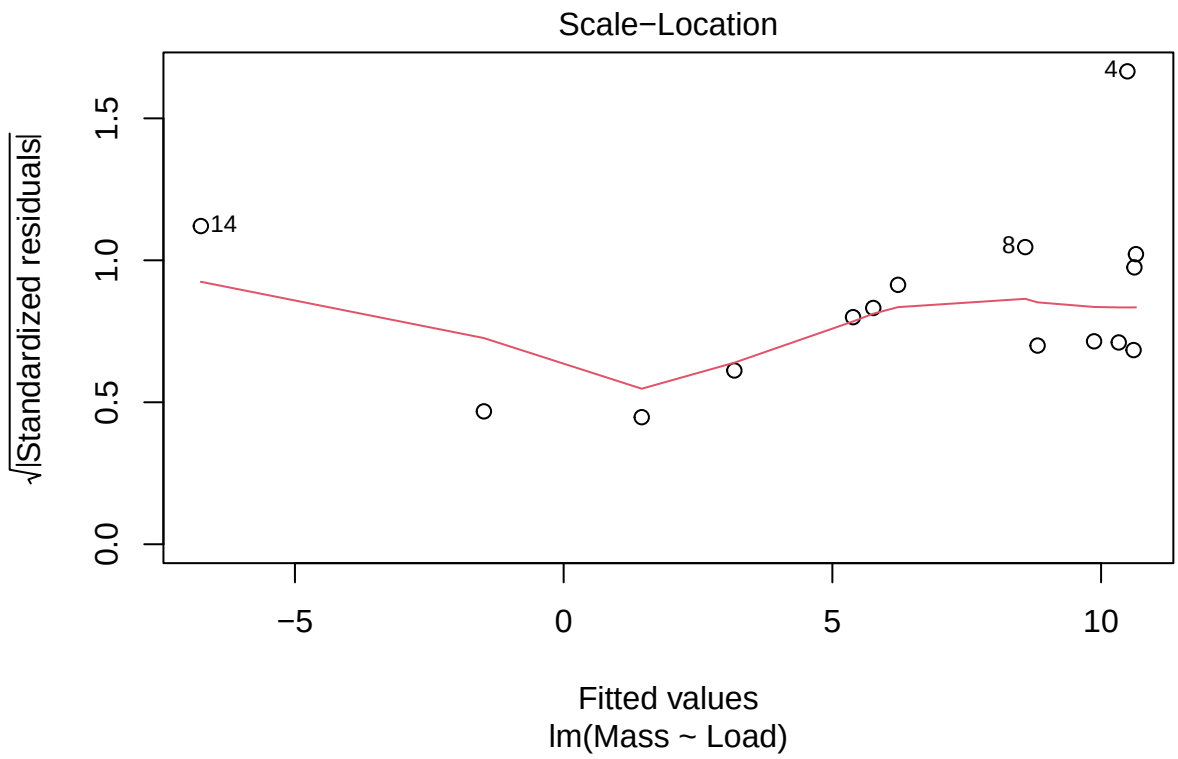


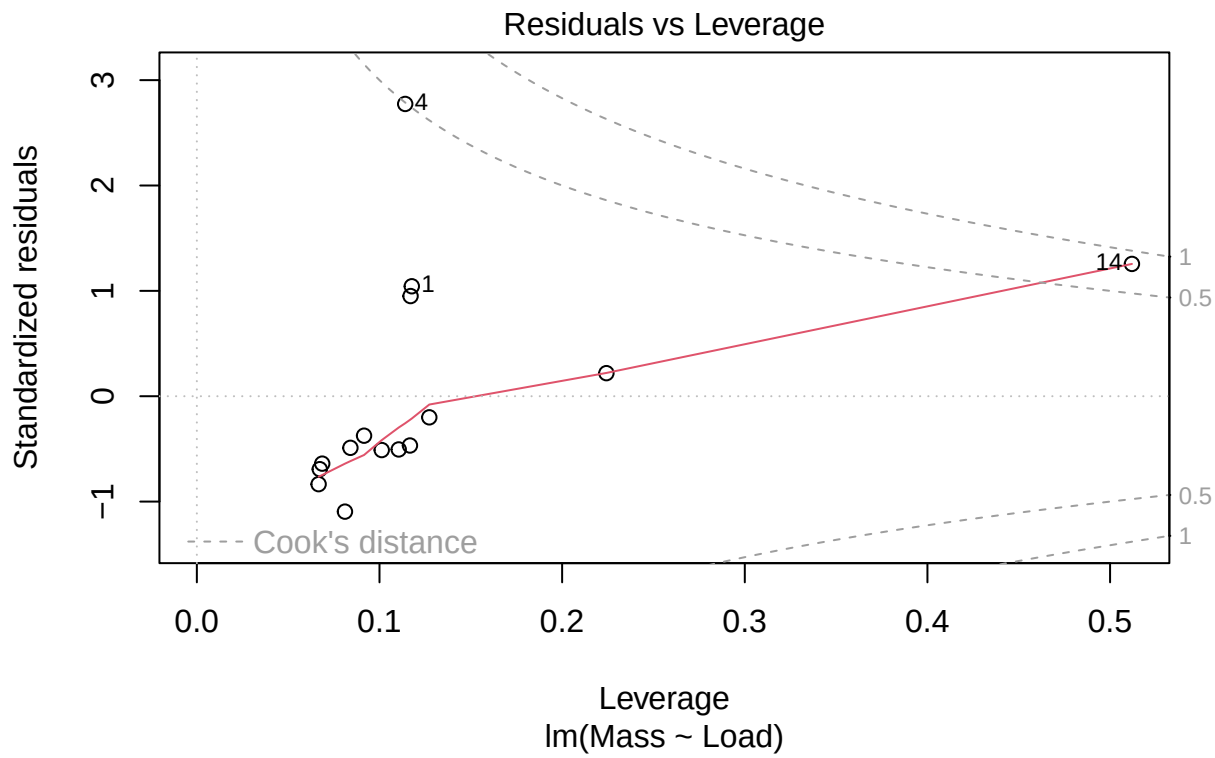
part b

```
flea_beetle <- lm(Mass ~ Load, data=ex0817)
plot(flea_beetle)
```


Residuals vs Fitted
Residuals
Fitted values
lm(Mass ~ Load)
4
1
8
20
15
10
5
0
−5
−10
−5
0
5
10

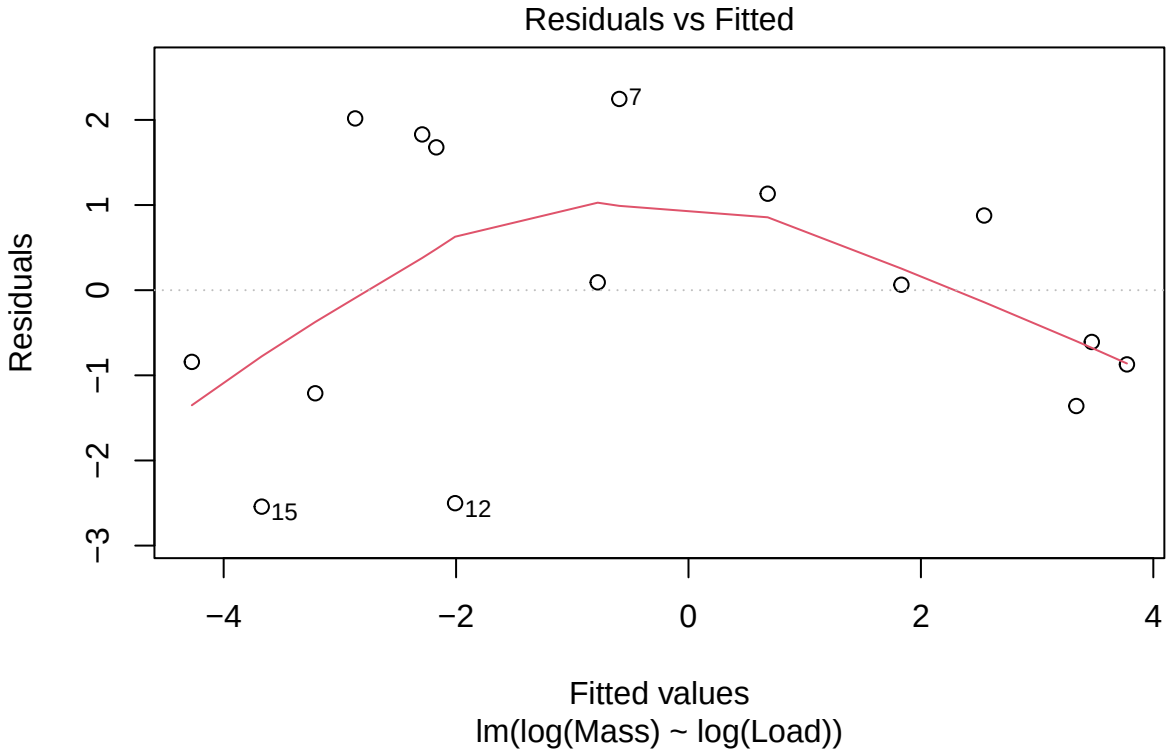

Q−Q Residuals
Standardized residuals
3
2
1
0
−1
4
14
8
−1
0
1
Theoretical Quantiles
lm(Mass ~ Load)

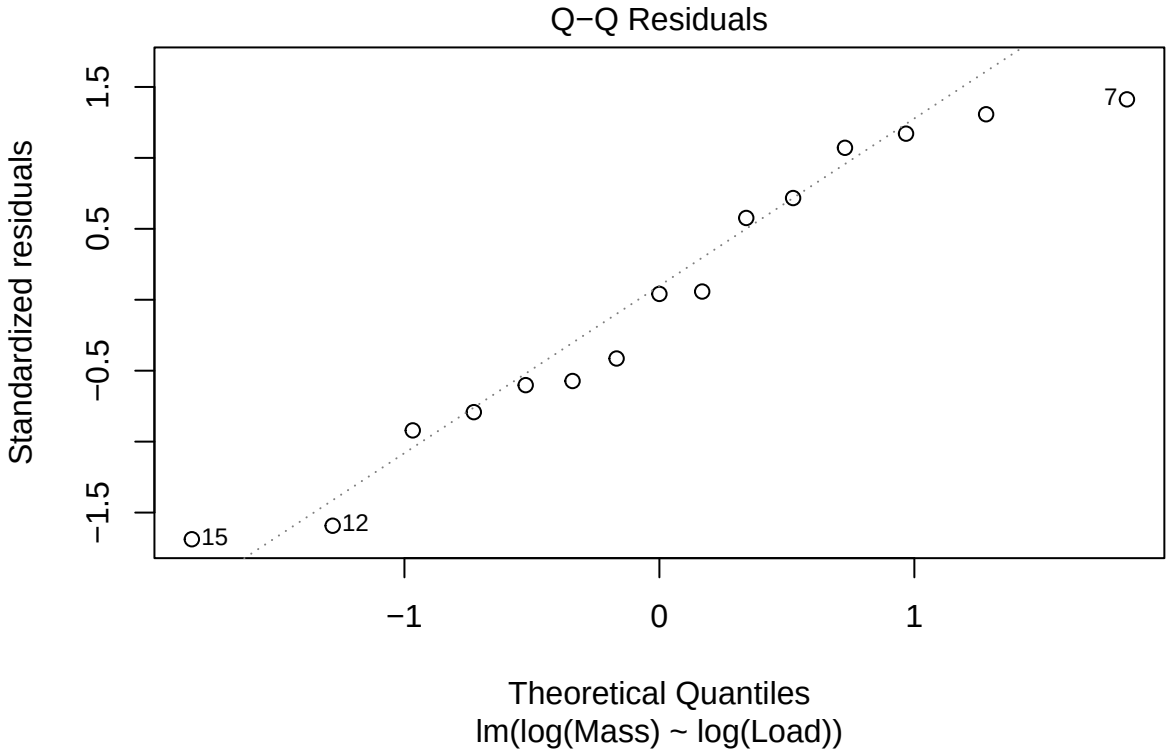

Scale−Location
√|Standardized residuals|
0.0
0.5
1.0
1.5
−5
0
5
10
14
8
4
Fitted values
lm(Mass ~ Load)

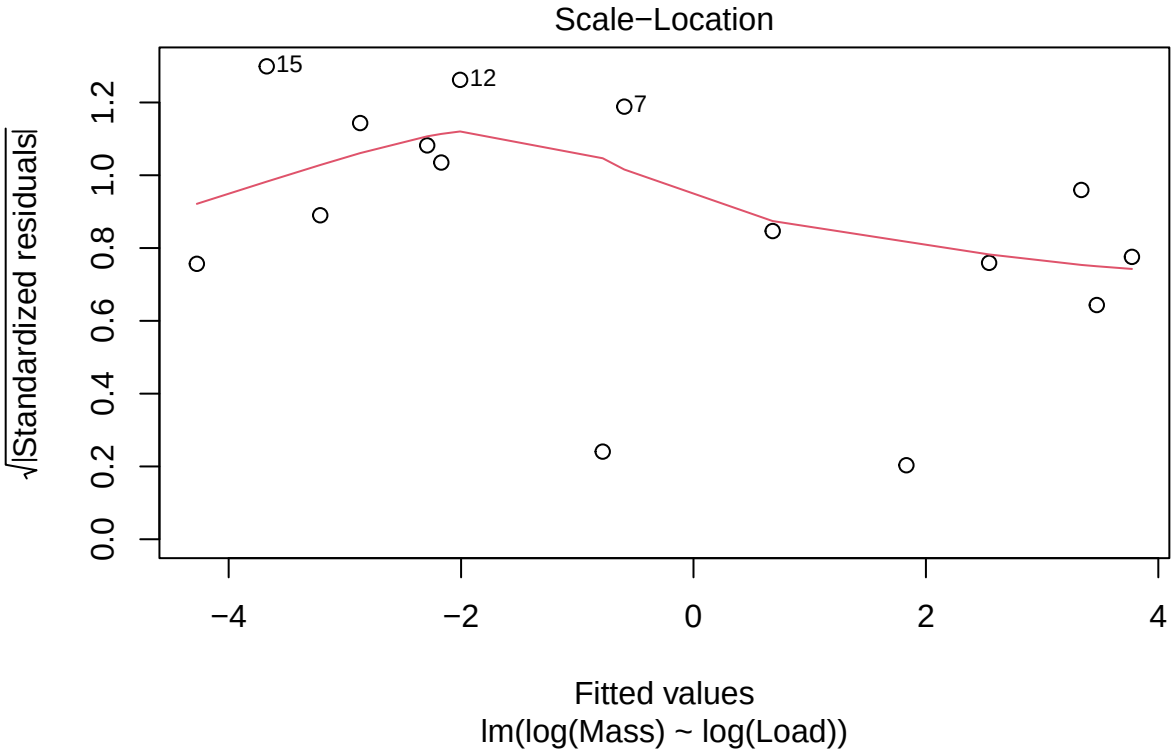


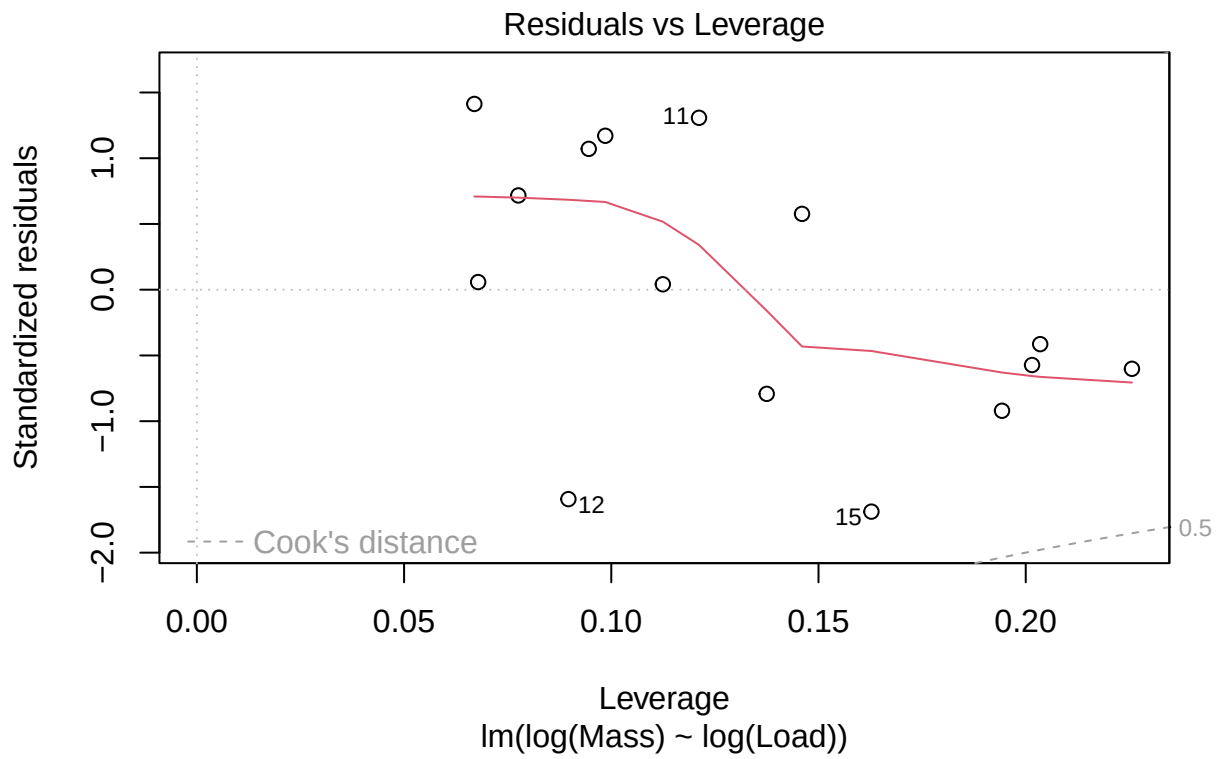
part c The residuals ~ Fitted values graph has an opposite Nike symbol thats downwards sloping. This could suggest a need for transformations, several examples of which are shown below.

```
log_beetles <- lm(log(Mass) ~ log(Load), data=ex0817)
plot(log_beetles)
```

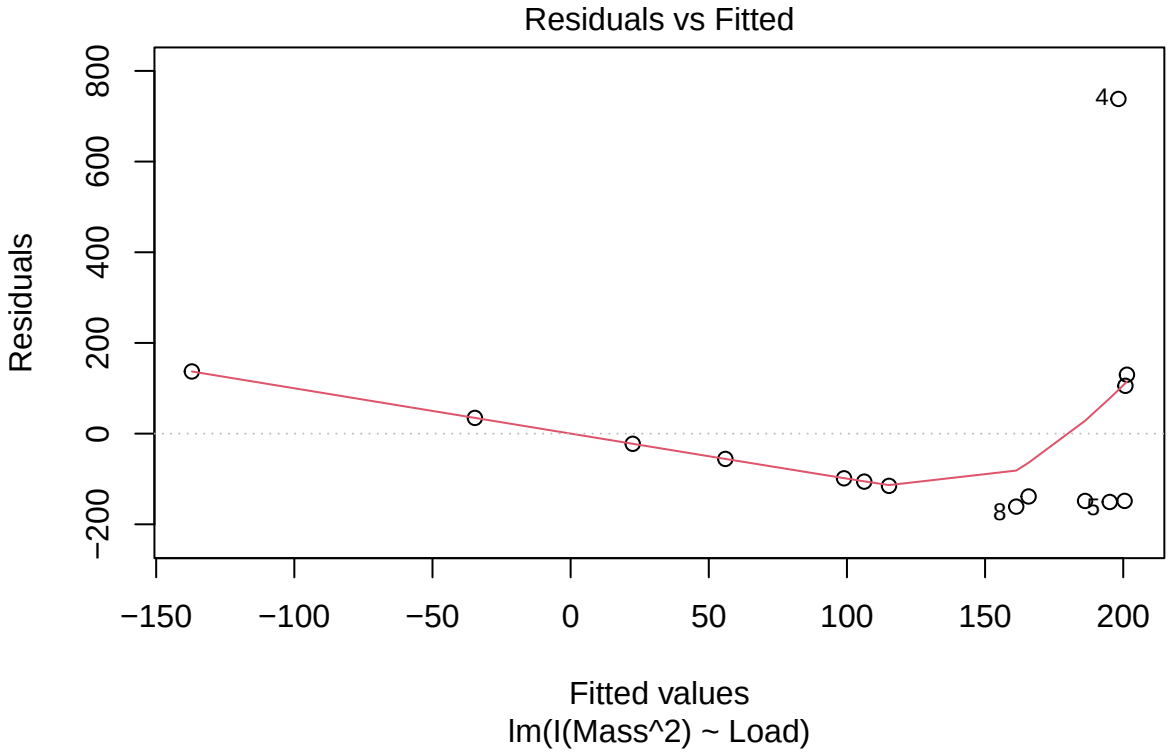

Residuals vs Fitted
Residuals
Fitted values
lm(log(Mass) ~ log(Load))
7
15
12
2
1
0
−1
−2
−3
−4
−2
0
2
4

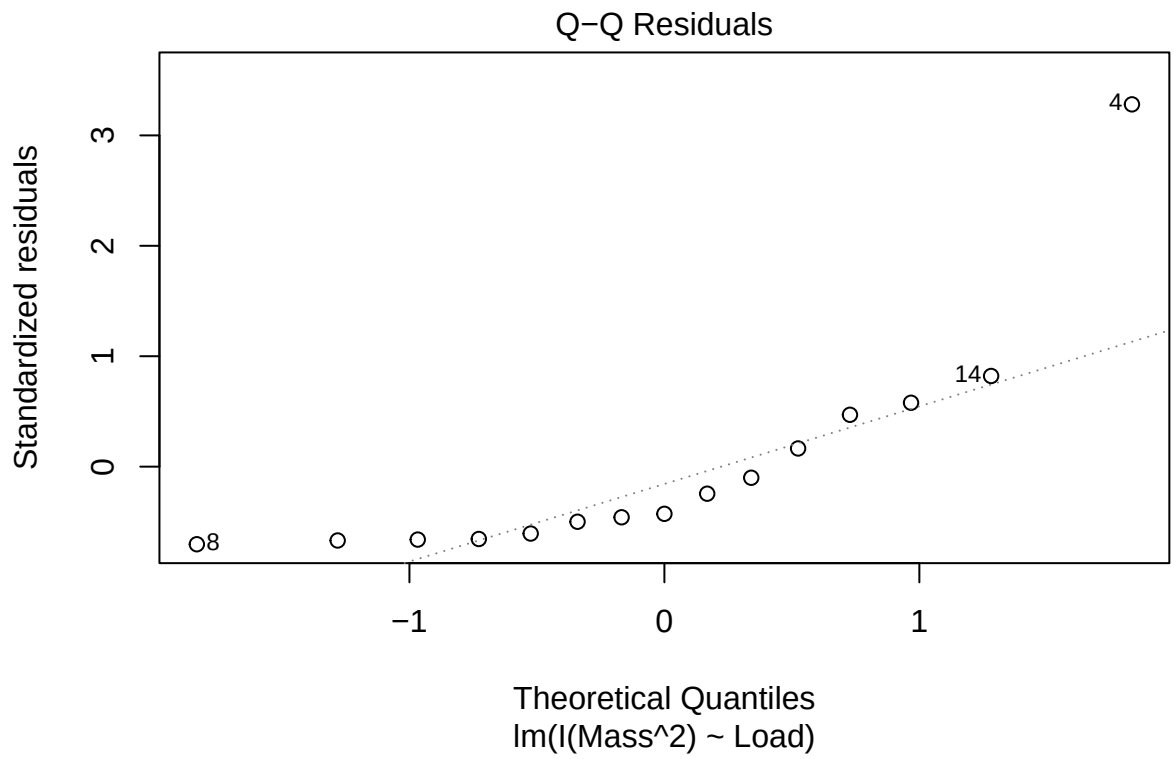

Q−Q Residuals
Standardized residuals
1.5
0.5
−0.5
−1.5
7
15
12
−1
0
1
Theoretical Quantiles
lm(log(Mass) ~ log(Load))

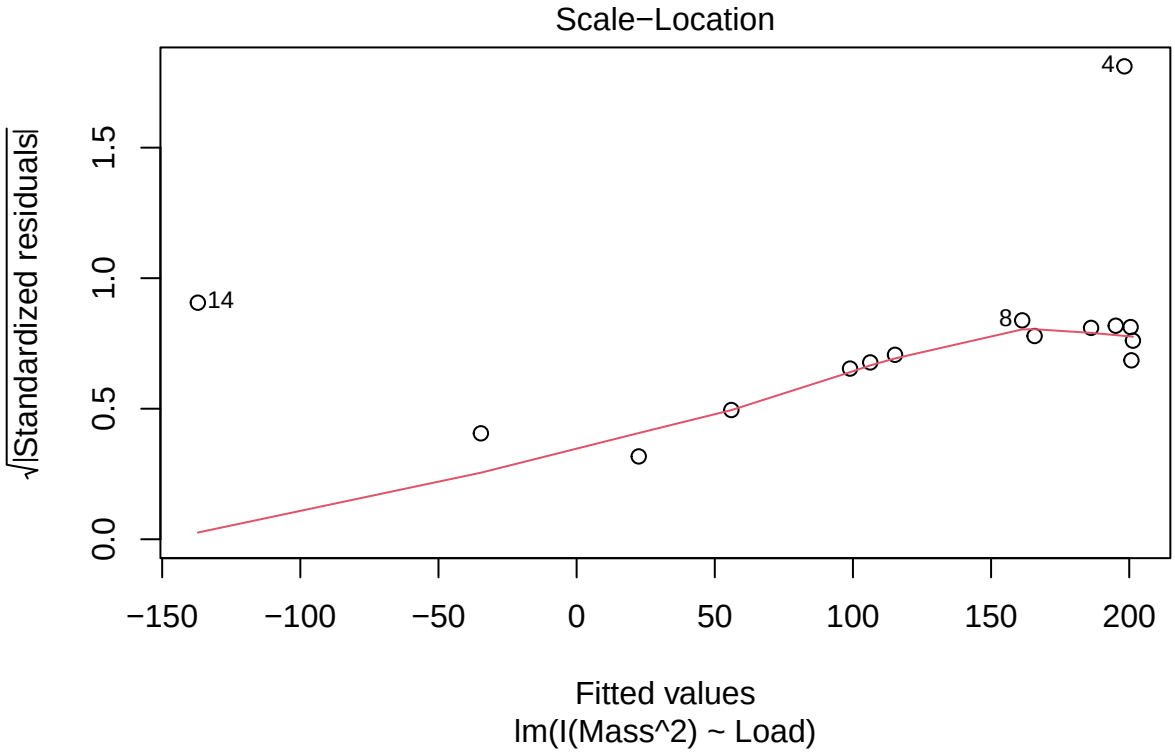

Scale−Location
15
12
7
√|Standardized residuals|
0.0
0.2
0.4
0.6
0.8
1.0
1.2
−4
−2
0
2
4
Fitted values
lm(log(Mass) ~ log(Load))

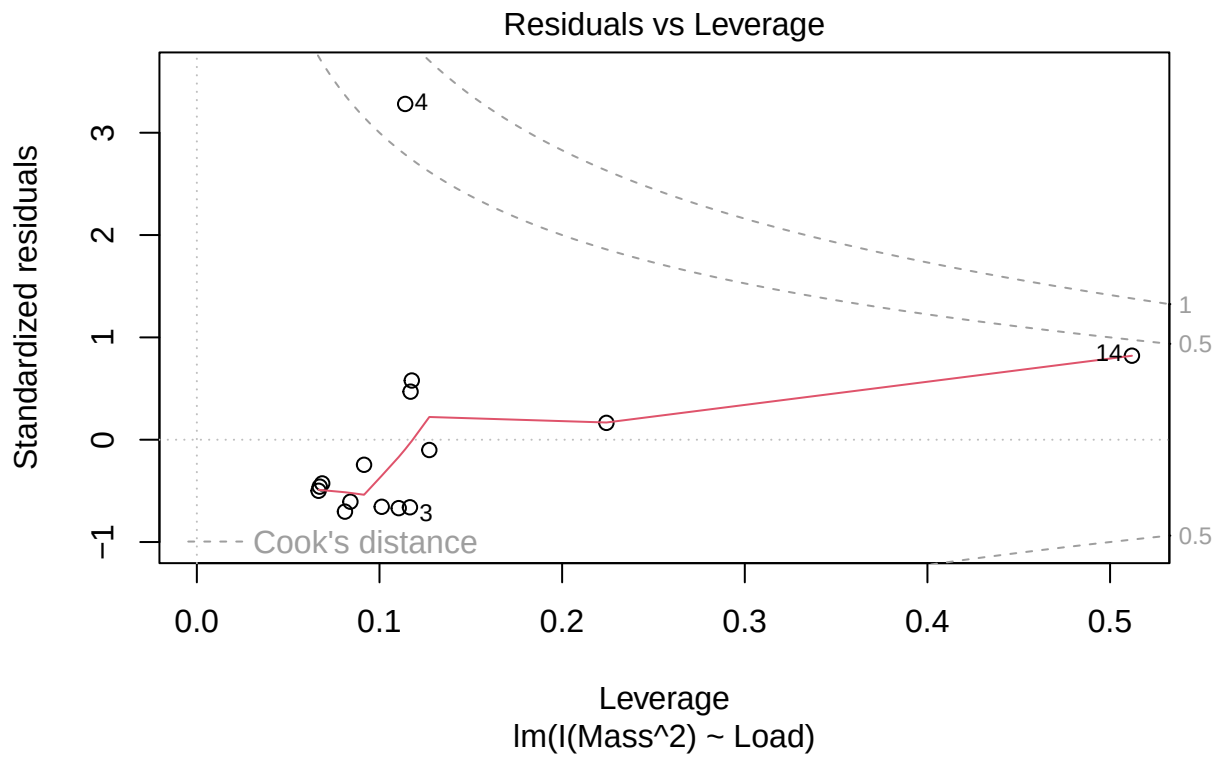



```
squared_beetles <- lm(I(Mass^2) ~ Load, data= ex0817)
plot(squared_beetles)
```

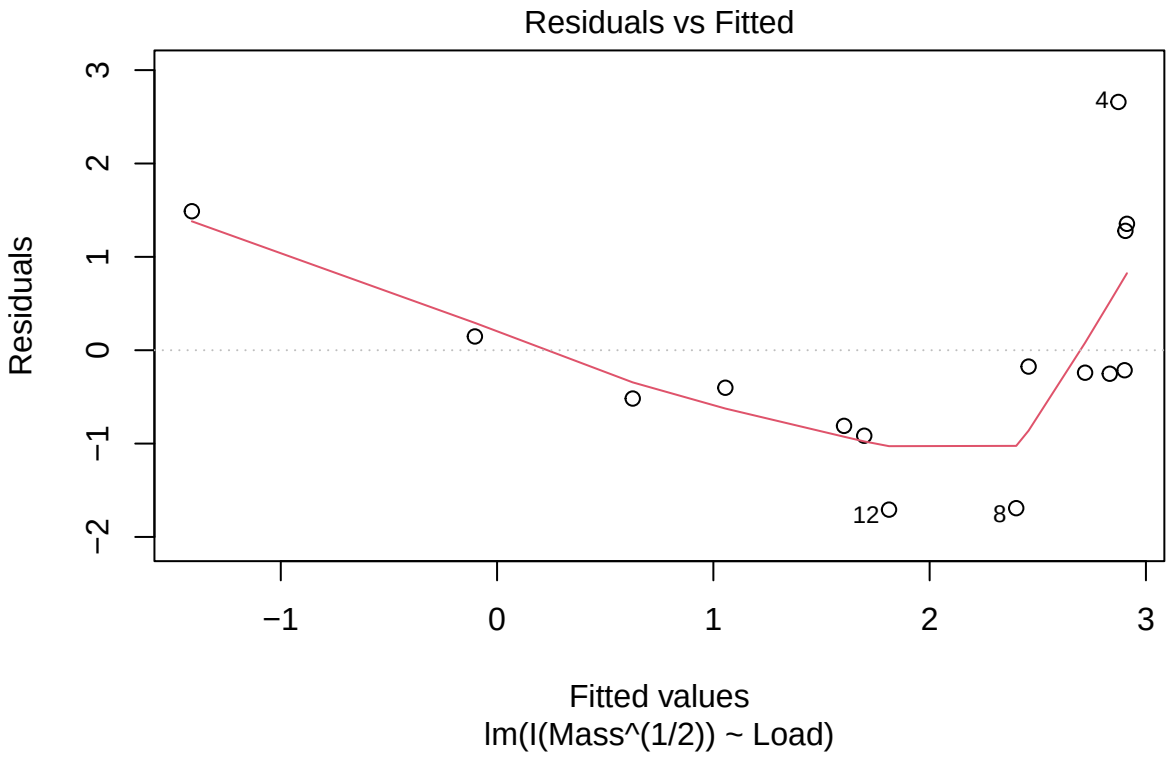

Residuals vs Fitted
Residuals
800
600
400
200
0
−200
4
8
5
−150
−100
−50
0
50
100
150
200
Fitted values
lm(I(Mass^2) ~ Load)

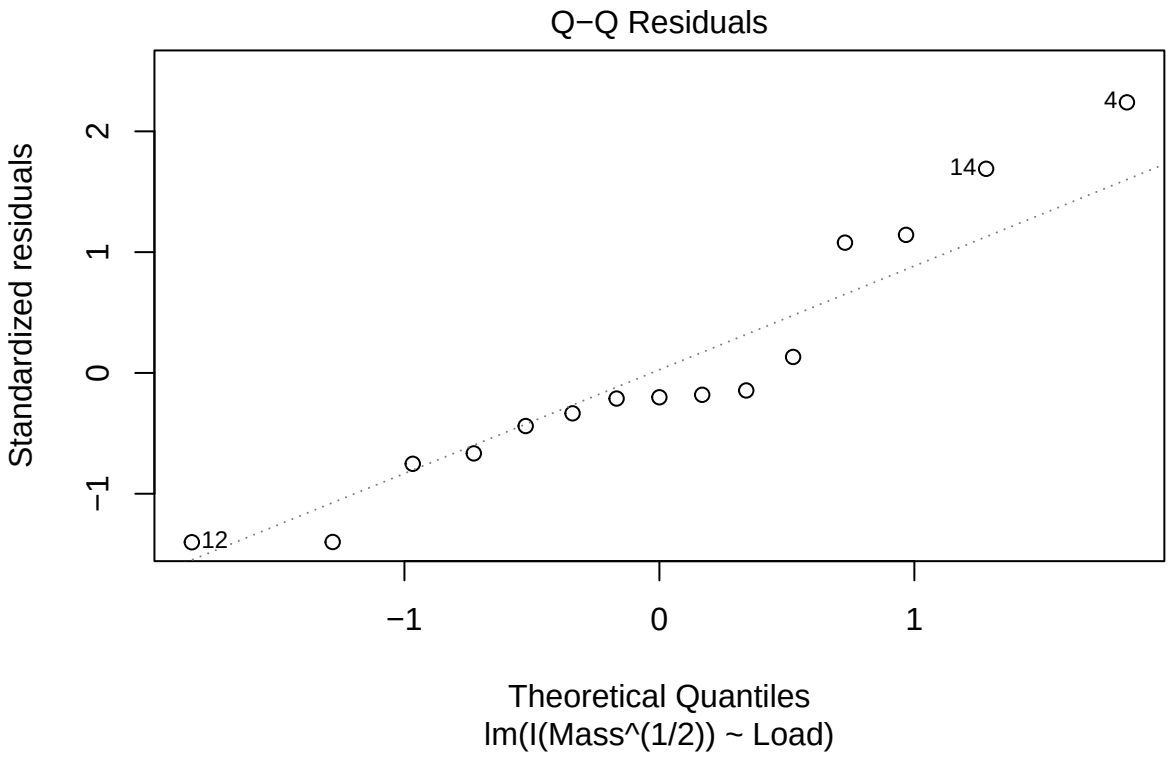

Q−Q Residuals
Standardized residuals
0
1
2
3
4
14
8
−1
0
1
Theoretical Quantiles
lm(I(Mass^2) ~ Load)

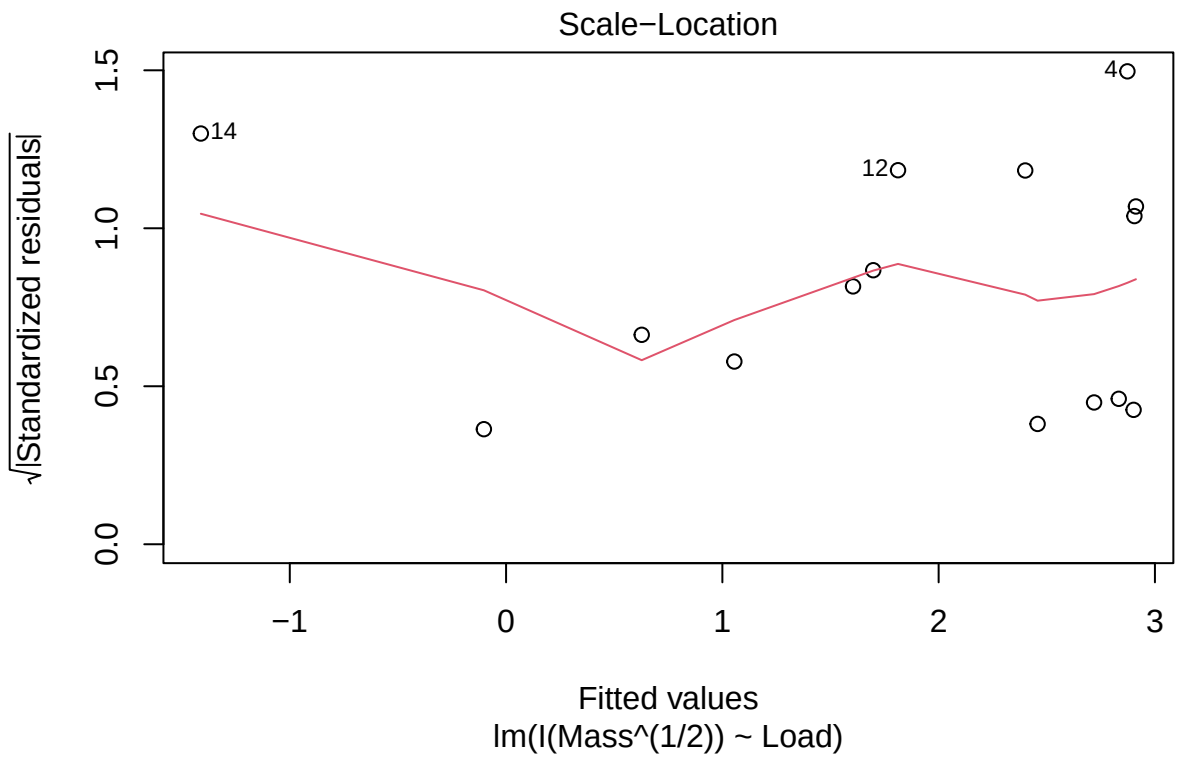

Scale−Location
√|Standardized residuals|
0.0
0.5
1.0
1.5
4
14
8
−150
−100
−50
0
50
100
150
200
Fitted values
lm(I(Mass^2) ~ Load)

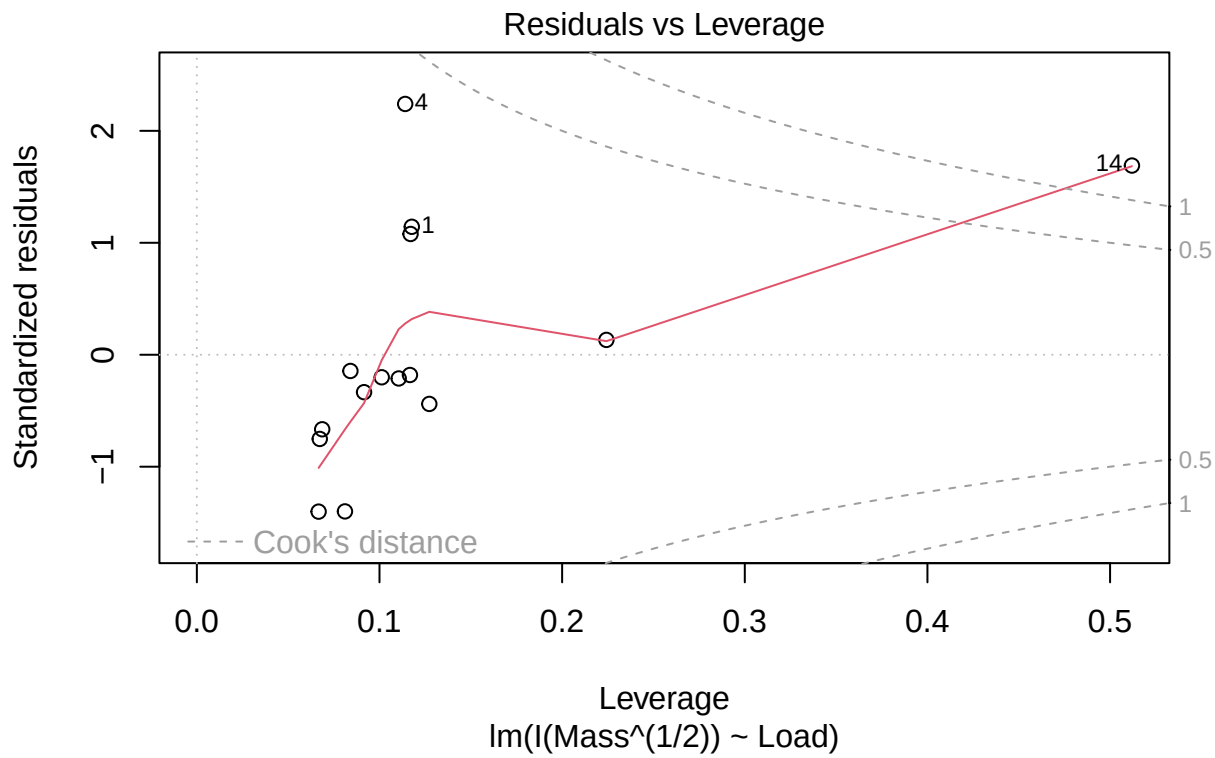



```r
sqrt_beetles <- lm(I(Mass^(1/2)) ~ Load, data = ex0817)
plot(sqrt_beetles)
```

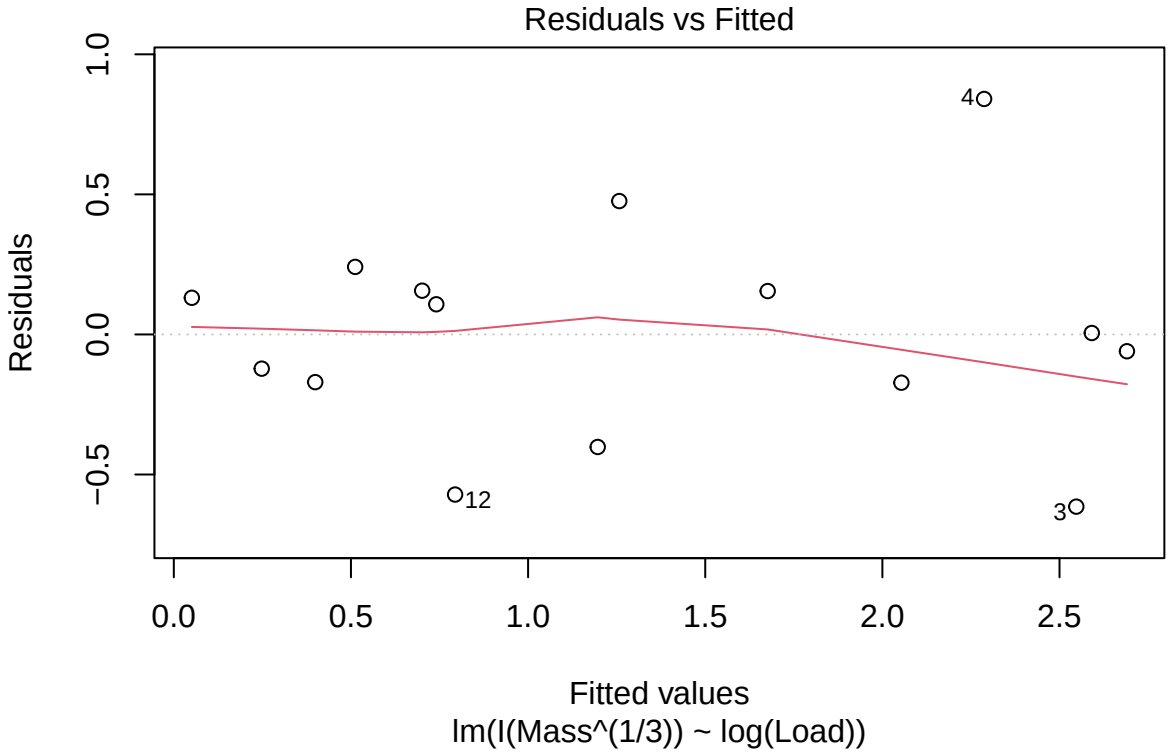

Residuals vs Fitted
Residuals
Fitted values
lm(I(Mass^(1/2)) ~ Load)
4
12
8
−2
−1
0
1
2
3

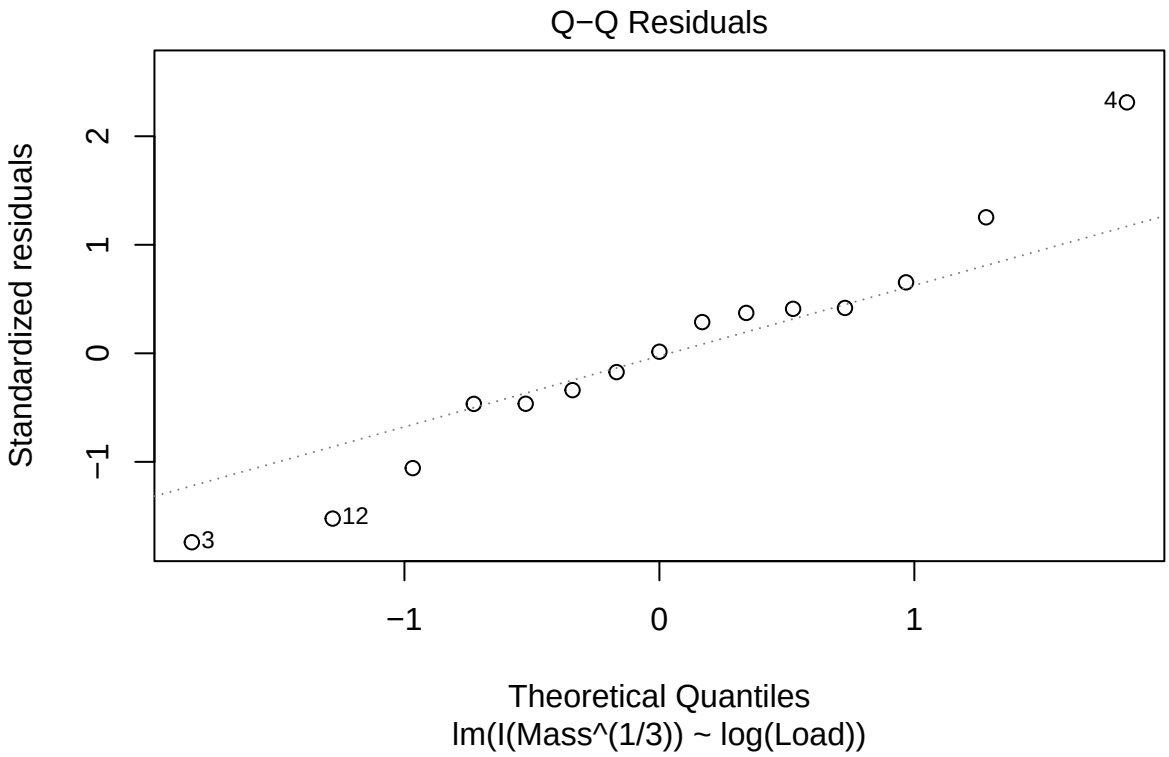

Q−Q Residuals
Standardized residuals
−1
0
1
2
4
14
12
−1
0
1
Theoretical Quantiles
lm(I(Mass^(1/2)) ~ Load)

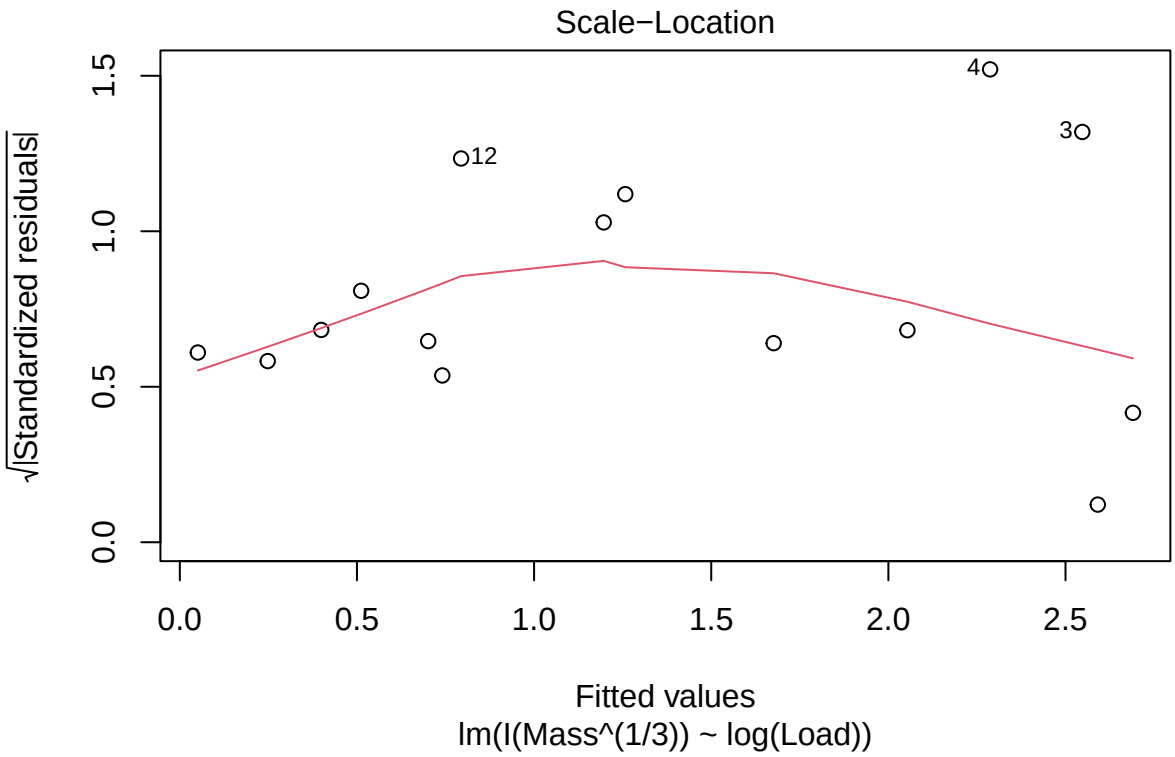

Scale−Location
√|Standardized residuals|
1.5
1.0
0.5
0.0
14
12
4
8
−1
0
1
2
3
Fitted values
lm(I(Mass^(1/2)) ~ Load)

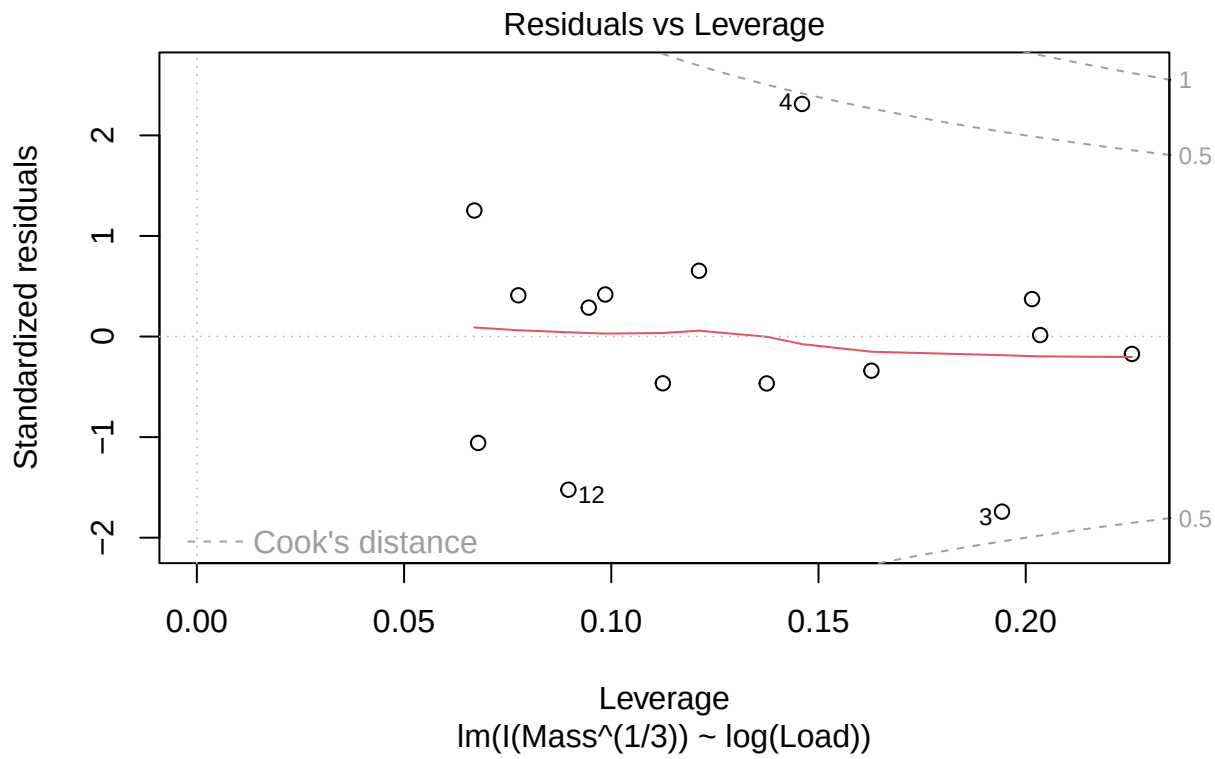



```r
cubedrt_beetles <- lm(I(Mass^(1/3)) ~ log(Load), data = ex0817)
plot(cubedrt_beetles)
```

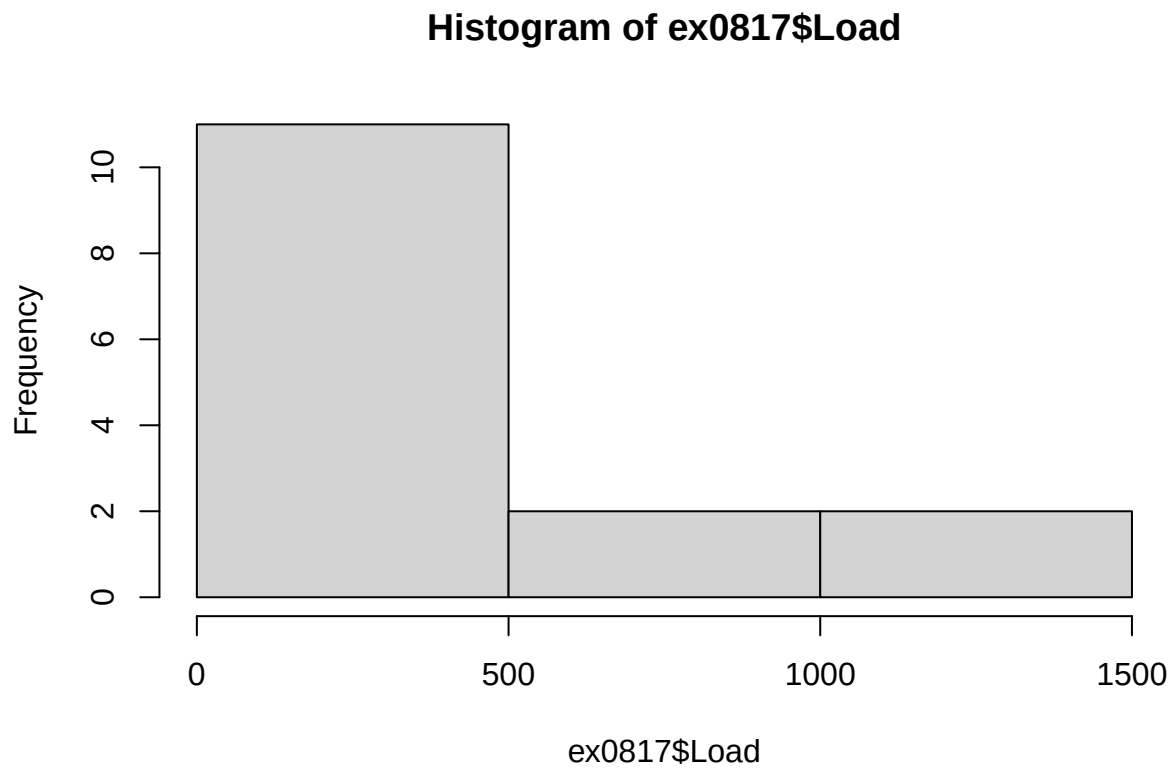

Residuals vs Fitted
Residuals
1.0
0.5
0.0
−0.5
4
12
3
0.0
0.5
1.0
1.5
2.0
2.5
Fitted values
lm(I(Mass^(1/3)) ~ log(Load))


Q−Q Residuals
Standardized residuals
2
1
0
−1
4
12
3
−1
0
1
Theoretical Quantiles
lm(I(Mass^(1/3)) ~ log(Load))


Scale−Location
√|Standardized residuals|
0.0
0.5
1.0
1.5
0.0
0.5
1.0
1.5
2.0
2.5
4
3
12
Fitted values
lm(I(Mass^(1/3)) ~ log(Load))



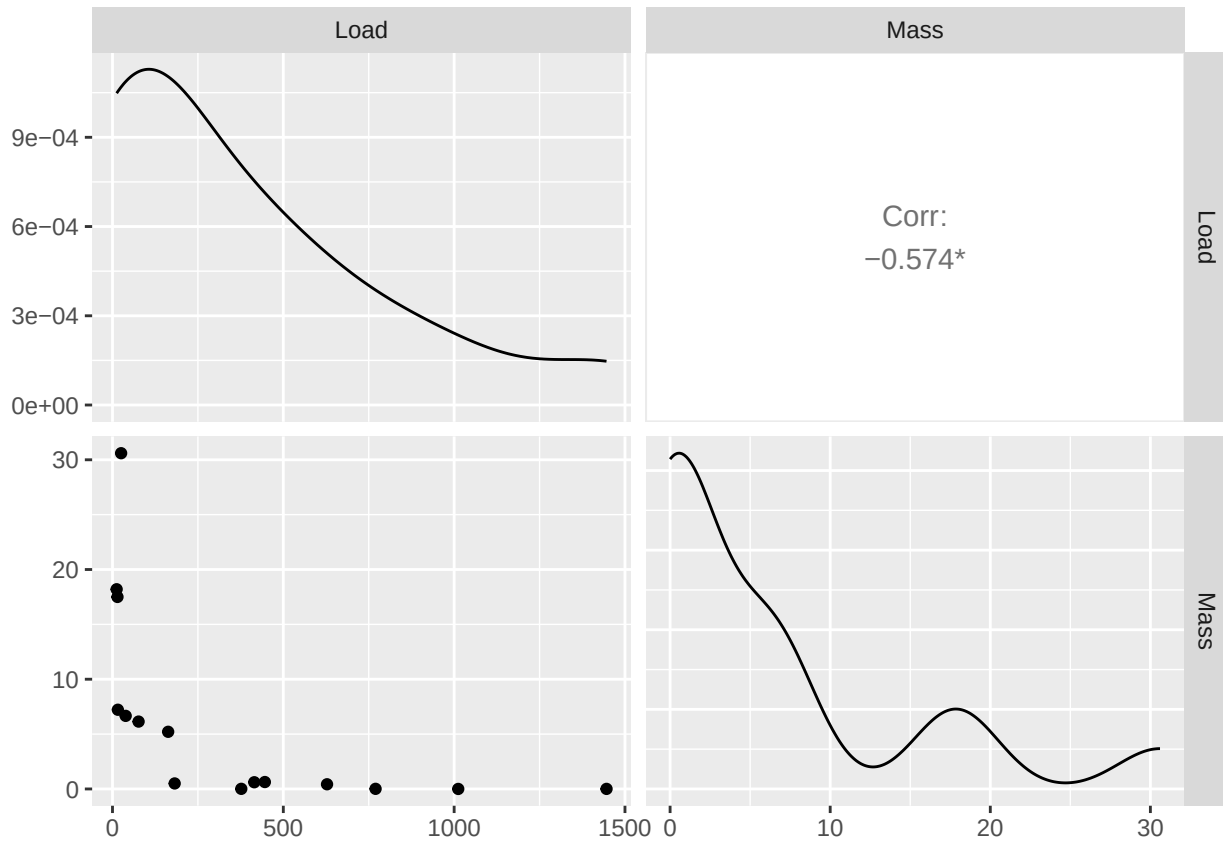

```r
hist(ex0817$Load)
```




```r
library(GGally)
```

```
## Registered S3 method overwritten by 'GGally':
##   method from
##   +.gg   ggplot2
```

```r
ex0817|> ggpairs()
```



Question 26

```
head(ex0826)
```

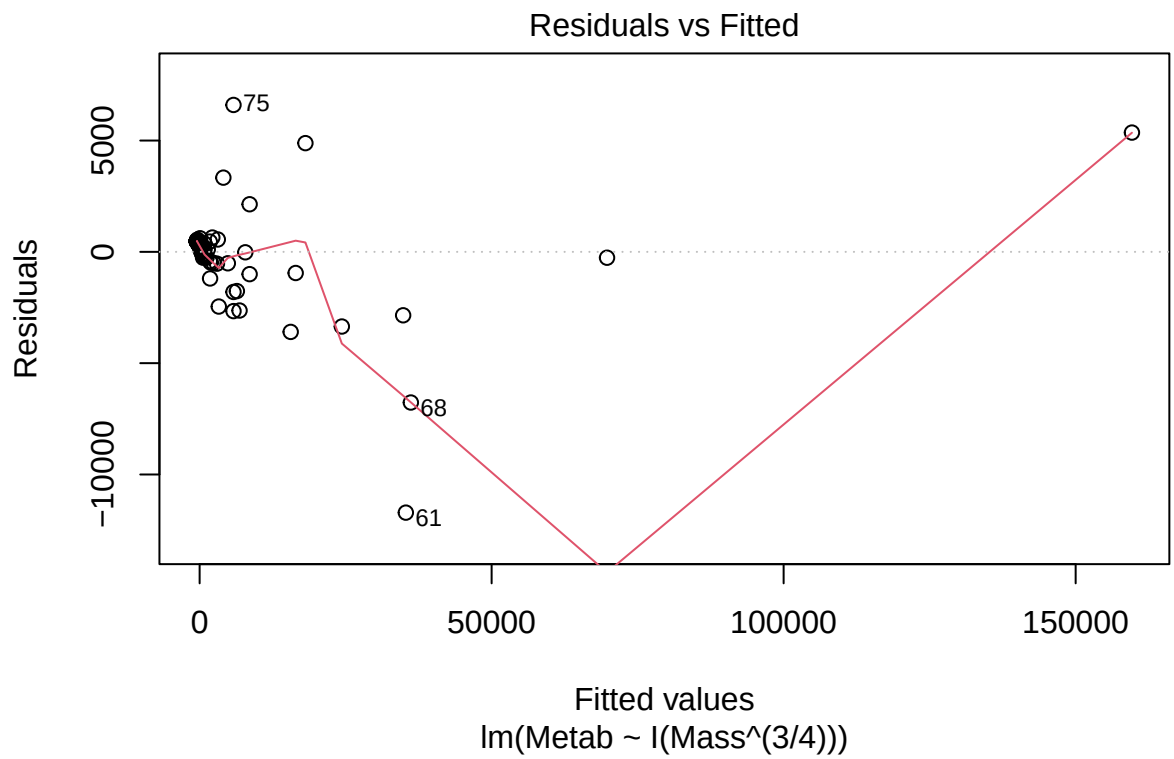
```
##                CommonName                   Species   Mass Metab Life
## 1                 Echidna   Tachiglossus aculeatus  2.500   302   14
## 2    Long-beaked echidna        Zaglossus bruijni 10.300   594   20
## 3                Platypus Ornithorhynchus anatinus  1.300   229    9
## 4                 Opossum Lutreolina crassicaudata  0.812   196    5
## 5 South American opossum    Didelphis marsupialis  1.330   299    6
## 6        Virginia opossum     Didelphis virginiana  3.260   519    8
```

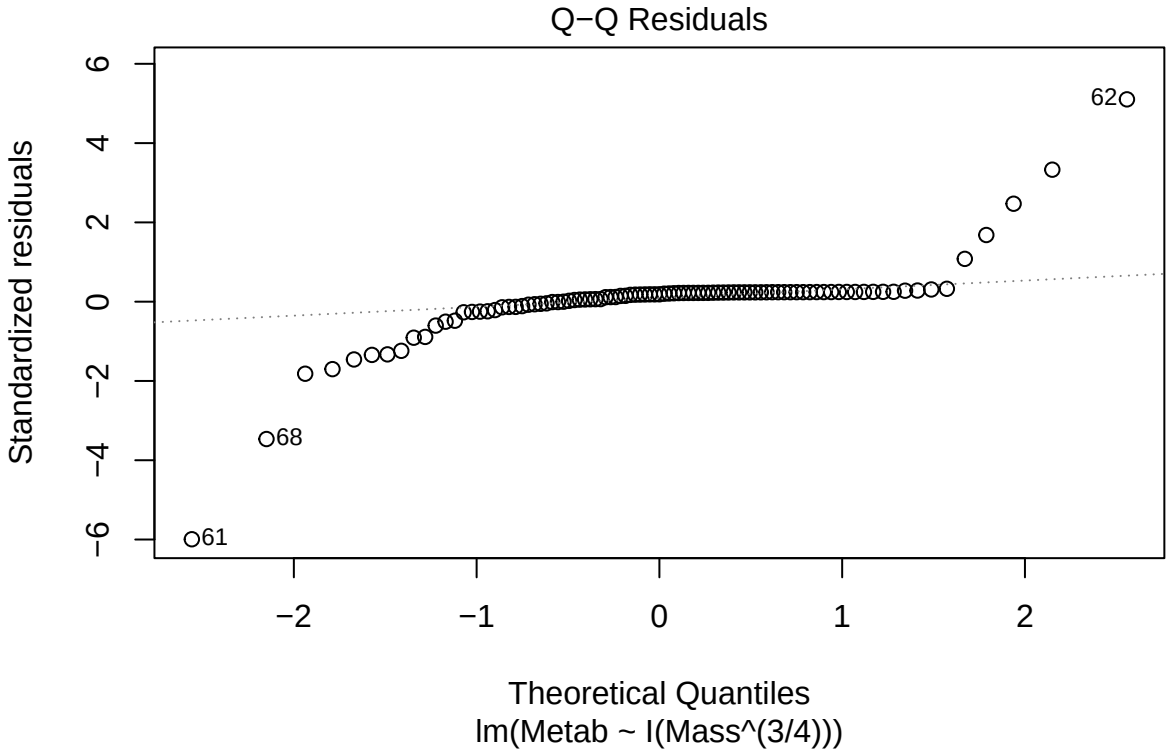
```
metabolic <- lm(Metab ~ I(Mass^(3/4)), data=ex0826)
summary(metabolic)
```

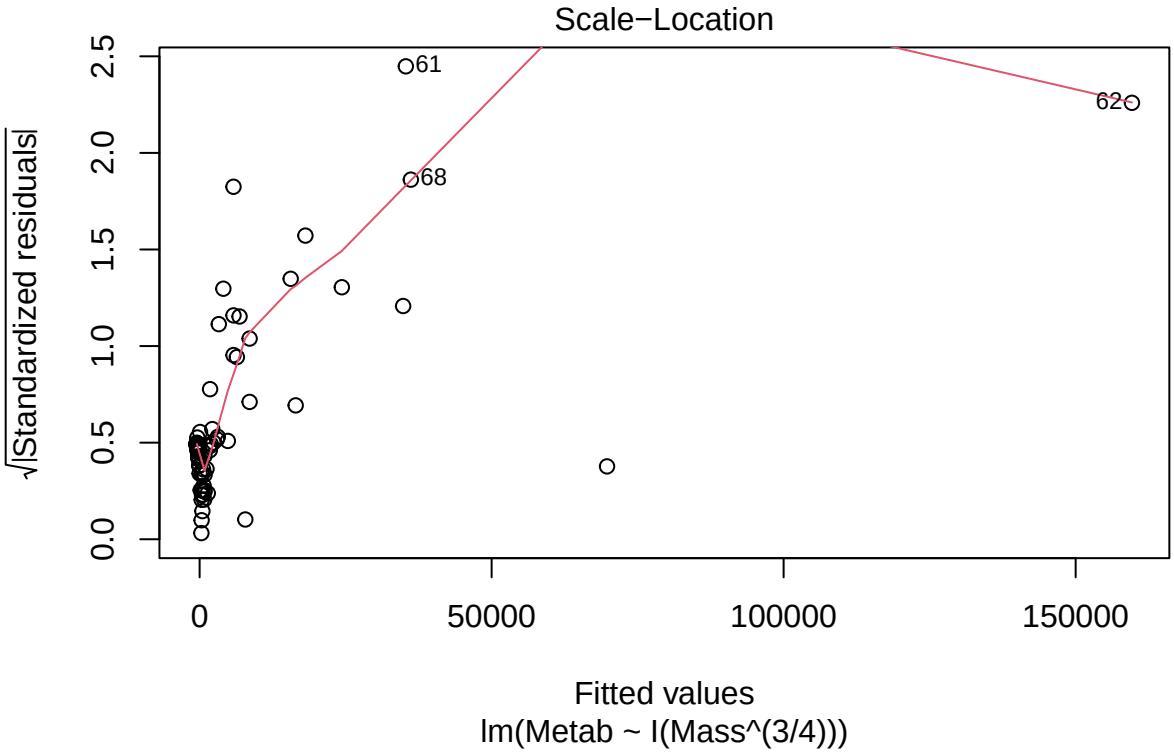
```
##
## Call:
## lm(formula = Metab ~ I(Mass^(3/4)), data = ex0826)
##
## Residuals:
##      Min       1Q   Median       3Q      Max
## -11712.7   -117.4    368.5    474.3   6598.5
##
```

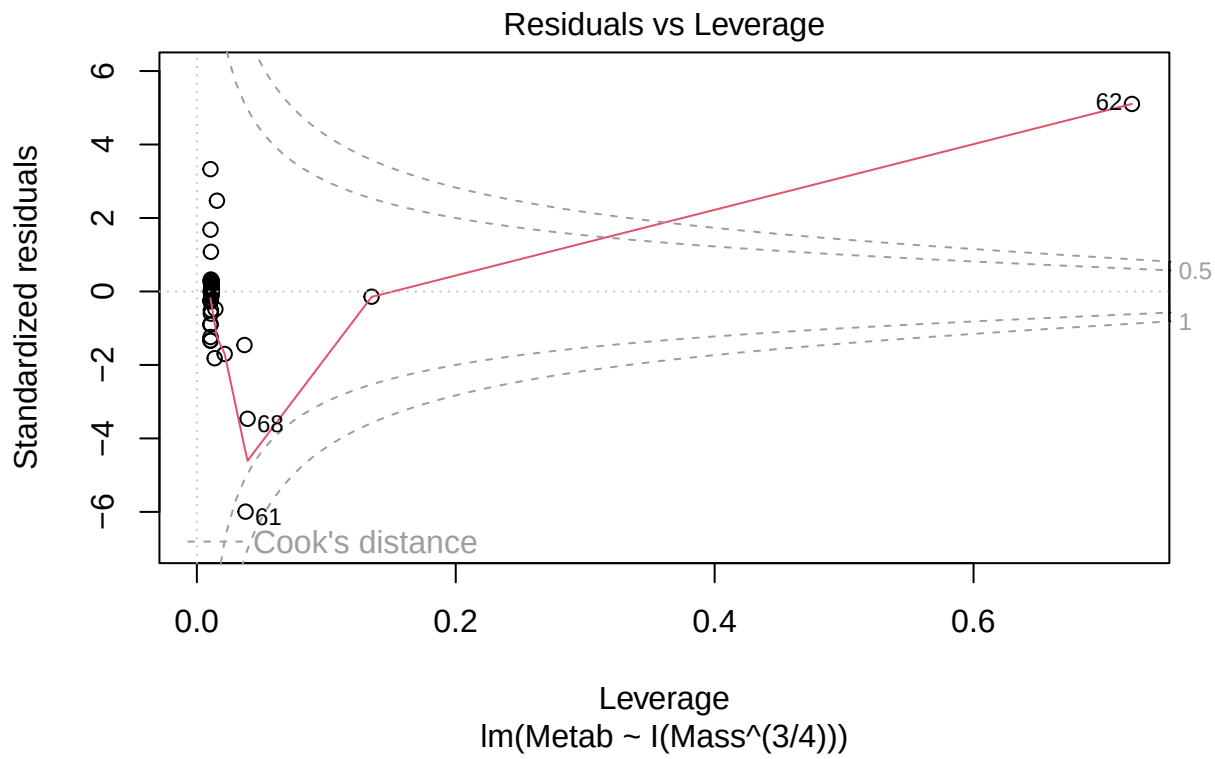
```
## Coefficients:
##                Estimate Std. Error t value Pr(>|t|)
## (Intercept)    -481.346    213.467  -2.255   0.0265 *
## I(Mass^(3/4))   395.016      4.299  91.895   <2e-16 ***
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
##
## Residual standard error: 1992 on 93 degrees of freedom
## Multiple R-squared:  0.9891, Adjusted R-squared:  0.989
## F-statistic:  8445 on 1 and 93 DF,  p-value: < 2.2e-16
```

```r
plot(metabolic)
```




Q−Q Residuals
Standardized residuals
6
4
2
0
−2
−4
−6
62
68
61
−2
−1
0
1
2
Theoretical Quantiles
lm(Metab ~ I(Mass^(3/4)))


Scale−Location
√|Standardized residuals|
0.0
0.5
1.0
1.5
2.0
2.5
61
62
68
0
50000
100000
150000
Fitted values
lm(Metab ~ I(Mass^(3/4)))



While I'm not totally confident I wrote Kleiber's Law relationship out correctly, if I did, I would not judge this theory as 100% accurate. While the relationship is significant, the plots show many possible concerns, such as leverage issues with an outlier, the residuals V fitted do not follow the straight line at all, and scale location map's line goes off the page.

Question 24

```r
hrt_rt <- lm(log(Rate) ~ Age, data = ex0824)
summary(hrt_rt)
```

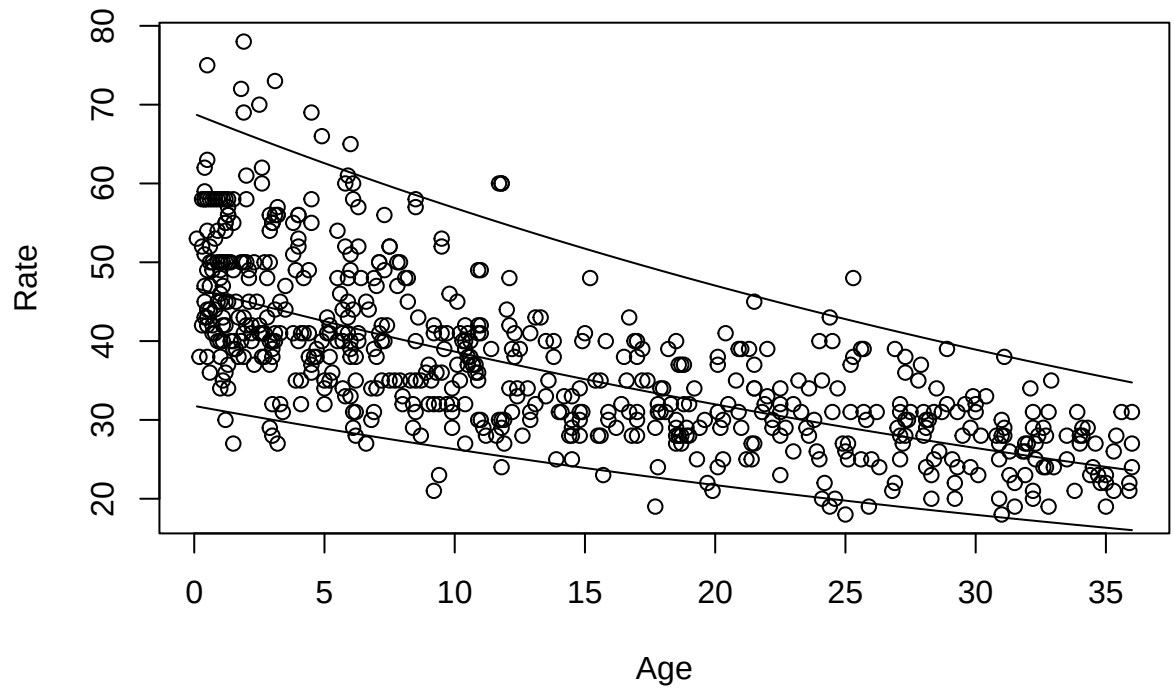
```
##
## Call:
## lm(formula = log(Rate) ~ Age, data = ex0824)
##
## Residuals:
##      Min       1Q   Median       3Q      Max
## -0.62571 -0.13201 -0.00402  0.13489  0.54771
##
## Coefficients:
##               Estimate Std. Error t value Pr(>|t|)
## (Intercept)  3.8451185  0.0126277  304.50   <2e-16 ***
## Age         -0.0190090  0.0007357  -25.84   <2e-16 ***
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
```

```
##
## Residual standard error: 0.1964 on 616 degrees of freedom
## Multiple R-squared:  0.5201, Adjusted R-squared:  0.5193
## F-statistic: 667.6 on 1 and 616 DF,  p-value: < 2.2e-16
```

```r
heart_pred <- predict(hrt_rt, interval= "prediction") |> exp()
```

```
## Warning in predict.lm(hrt_rt, interval = "prediction"): predictions on current data refer to _future_
```

```r
plot_data <- cbind(ex0824, heart_pred)

plot(Rate ~ Age, data = ex0824)

lines(plot_data$Age, plot_data$fit)

lines(plot_data$Age, plot_data$lwr)
lines(plot_data$Age, plot_data$upr)
```

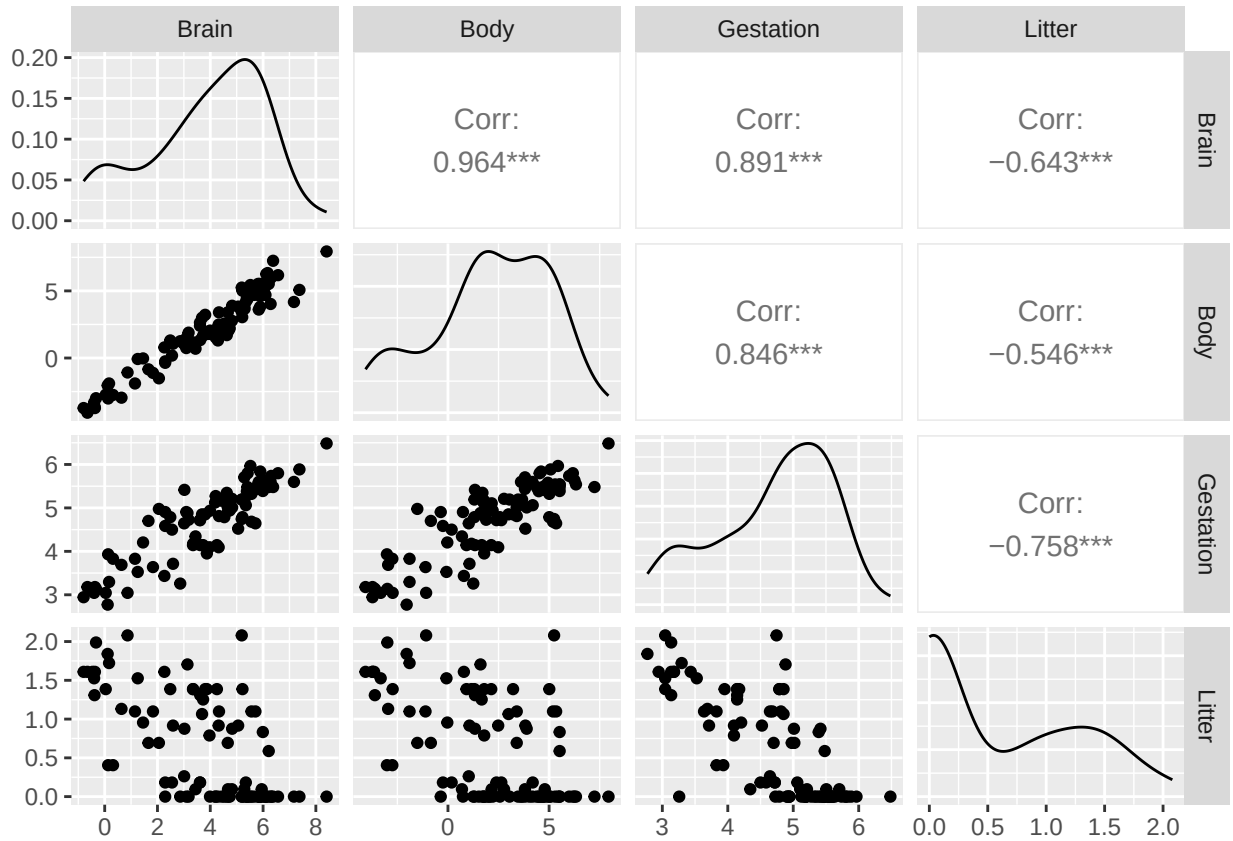


Chapter 9

Question 12

part a

```r
library(GGally)

log_mammals <- case0902 |>
  select(Brain, Body, Gestation, Litter) |>
  mutate(across(everything(), log))

ggpairs(log_mammals)
```



part b

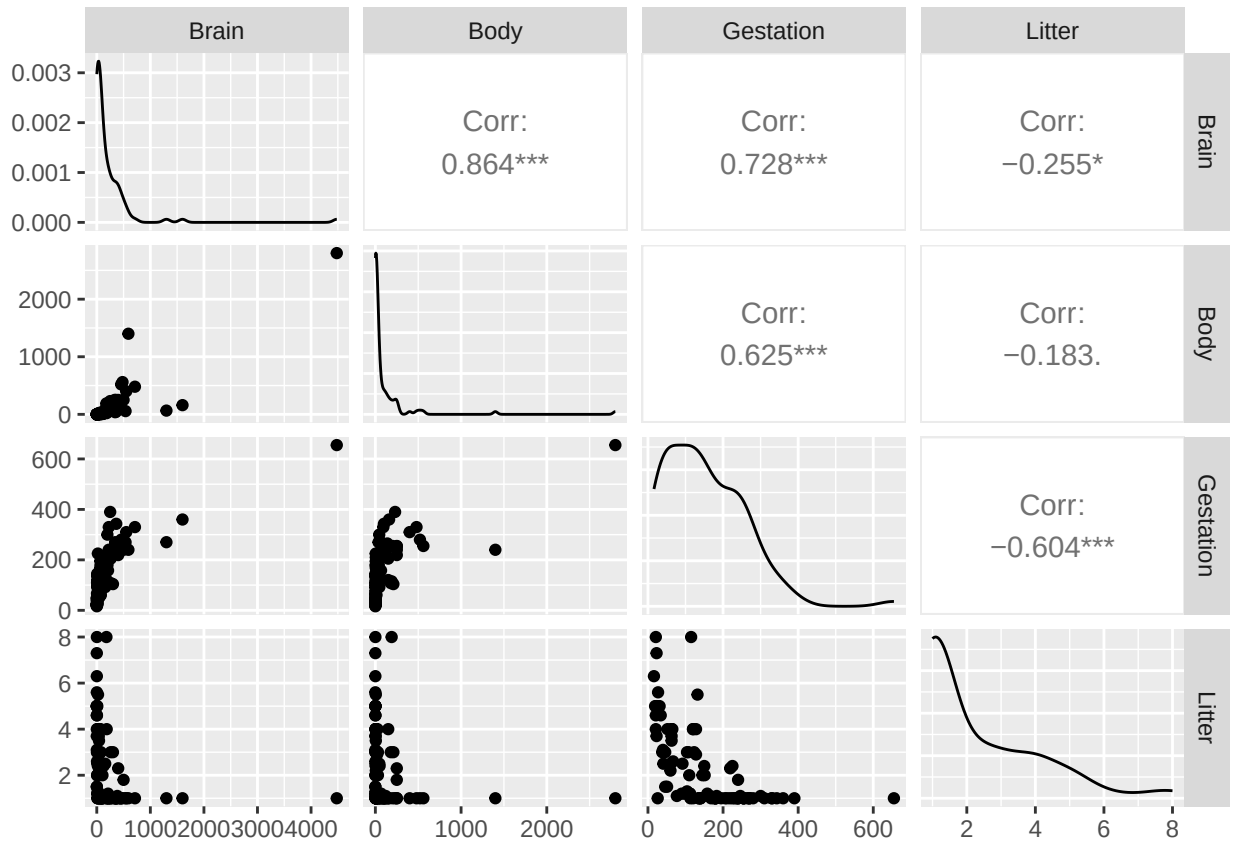
```r
log_brain_weight <- lm(log(Brain) ~ log(Body) + log(Gestation) + log(Litter), data = case0902)
summary(log_brain_weight)
```

```
##
## Call:
## lm(formula = log(Brain) ~ log(Body) + log(Gestation) + log(Litter),
##     data = case0902)
##
## Residuals:
##      Min       1Q   Median       3Q      Max
## -0.95415 -0.29639 -0.03105  0.28111  1.57491
##
## Coefficients:
##                Estimate Std. Error t value Pr(>|t|)
## (Intercept)     0.85482    0.66167   1.292  0.19962
```

```
## log(Body)       0.57507    0.03259  17.647  < 2e-16 ***
## log(Gestation)  0.41794    0.14078   2.969  0.00381 **
## log(Litter)    -0.31007    0.11593  -2.675  0.00885 **
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
##
## Residual standard error: 0.4748 on 92 degrees of freedom
## Multiple R-squared:  0.9537, Adjusted R-squared:  0.9522
## F-statistic: 631.6 on 3 and 92 DF,  p-value: < 2.2e-16
```

part c

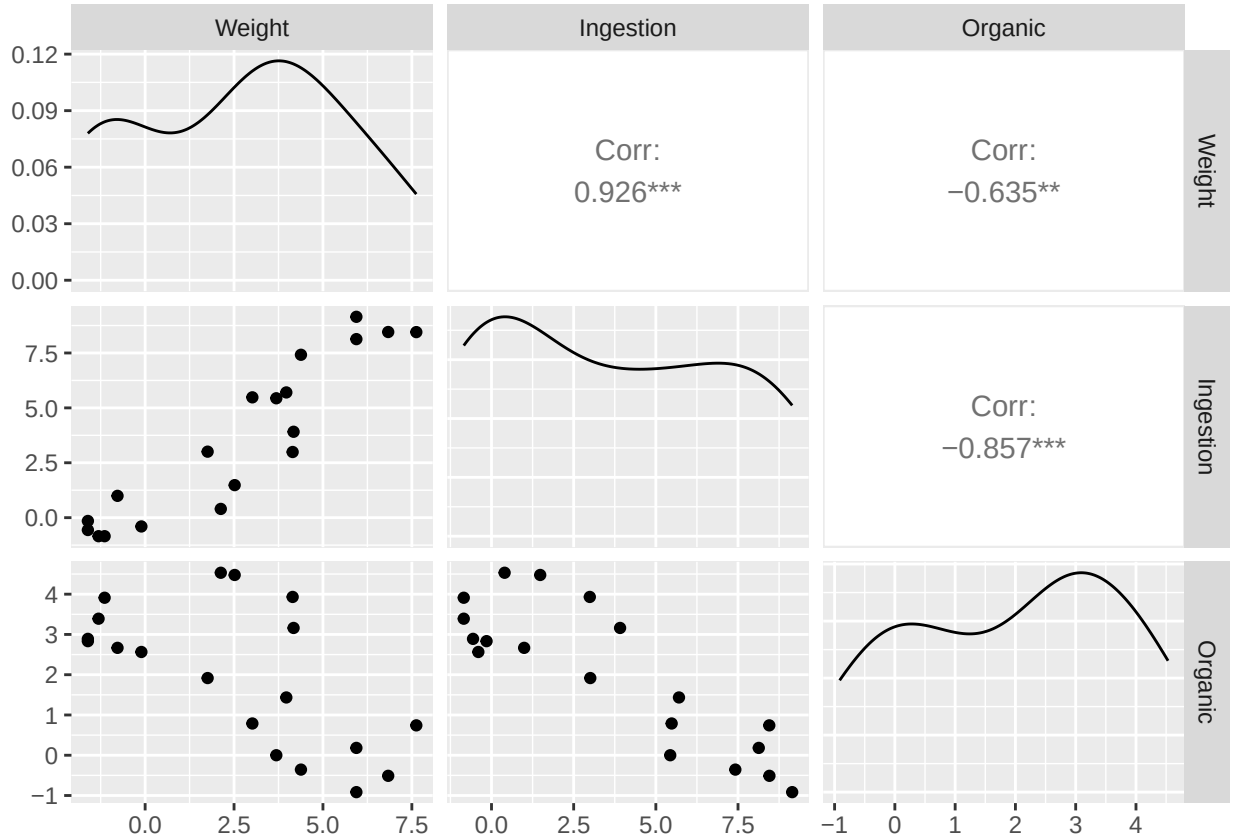
```r
nolog_mammals <- case0902 |>
  select(Brain, Body, Gestation, Litter) |>
  mutate(across(everything()))

ggpairs(nolog_mammals)
```



The linearity of the log transformed data looks much better than the original data, which is simply because there are so many of the data observances in the smaller ranges, which distorts the visual spread of the data's relationships.

Question 21

```
log_case <- ex0921 |>
  select(Weight, Ingestion, Organic) |>
  mutate(across(everything(), log))
ggpairs(log_case)
```




```
log_weight_organic <- lm(log(Ingestion) ~ log(Weight) + log(Organic), data= ex0921)
```

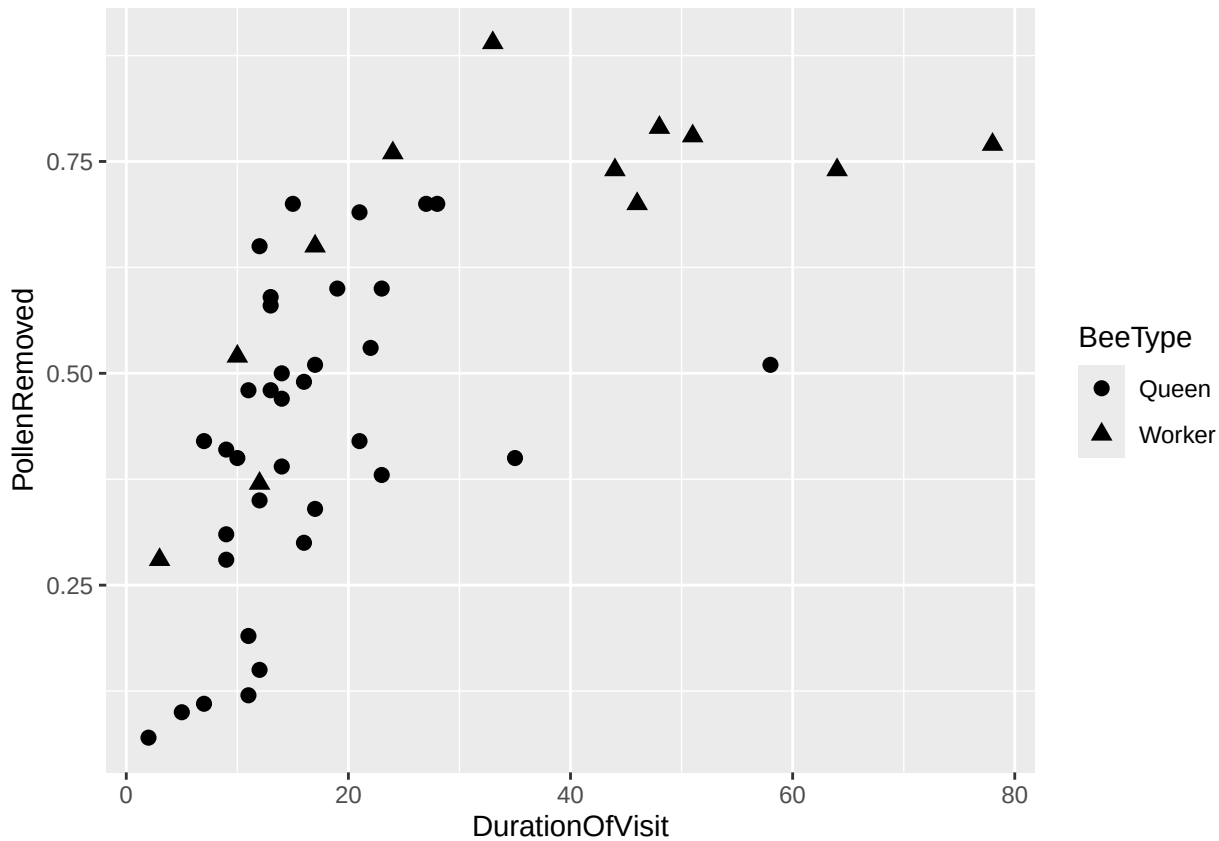
Question 16

part a

```
head(ex0327)
```

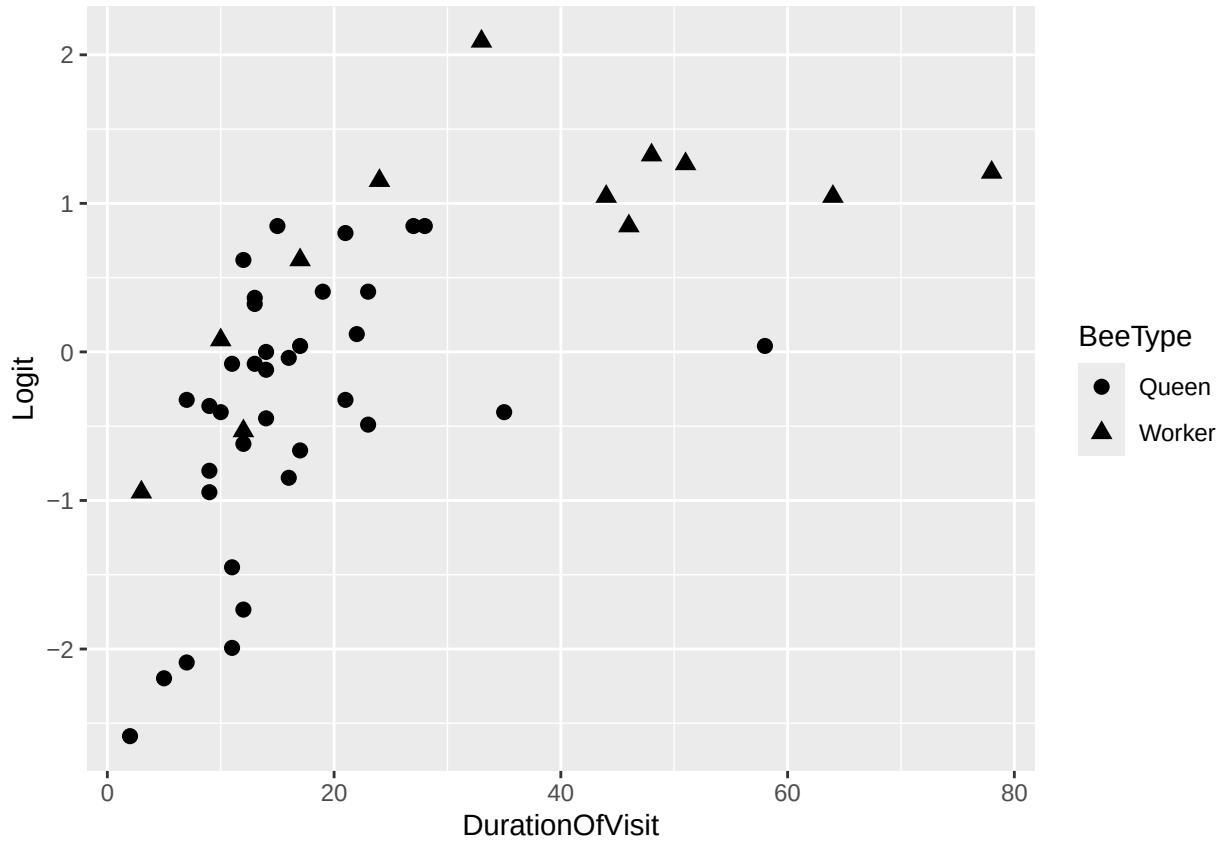
```
##   PollenRemoved DurationOfVisit BeeType
## 1          0.07               2   Queen
## 2          0.10               5   Queen
## 3          0.11               7   Queen
## 4          0.12              11   Queen
## 5          0.15              12   Queen
## 6          0.19              11   Queen
```

```r
ggplot(ex0327, aes(x = DurationOfVisit, y = PollenRemoved, shape = BeeType)) +
  geom_point(size = 2.5)
```



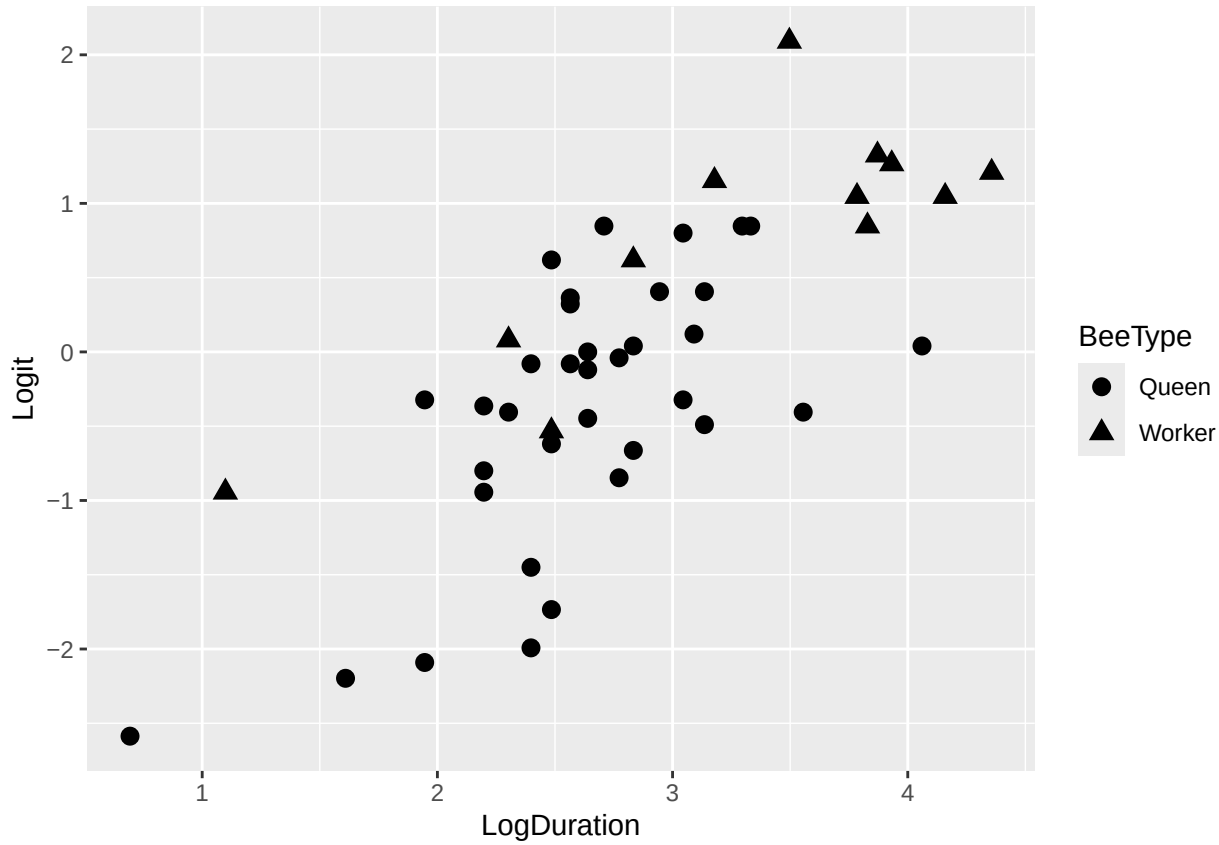
part b

```r
Logit <- qlogis(ex0327$PollenRemoved)

ggplot(ex0327, aes(x = DurationOfVisit, y = Logit, shape = BeeType)) +
  geom_point(size = 2.5)
```



part c

```
LogDuration <- log(ex0327$DurationOfVisit)

ggplot(ex0327, aes(x = LogDuration, y = Logit, shape = BeeType)) +
  geom_point(size = 3)
```



part d

```
Bee <- factor(ex0327$BeeType)

model_interact <- lm(Logit ~ LogDuration * Bee, data = ex0327)
summary(model_interact)
```

```
##
## Call:
## lm(formula = Logit ~ LogDuration * Bee, data = ex0327)
##
## Residuals:
##     Min      1Q  Median      3Q     Max
## -1.3803 -0.3699  0.0307  0.4552  1.1611
##
## Coefficients:
##                       Estimate Std. Error t value Pr(>|t|)
## (Intercept)            -3.0390     0.5115  -5.941 4.45e-07 ***
## LogDuration             1.0121     0.1902   5.321 3.52e-06 ***
## BeeWorker               1.3770     0.8722   1.579    0.122
## LogDuration:BeeWorker  -0.2709     0.2817  -0.962    0.342
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
##
## Residual standard error: 0.6525 on 43 degrees of freedom
## Multiple R-squared:  0.6151, Adjusted R-squared:  0.5882
```

```
## F-statistic:  22.9 on 3 and 43 DF,  p-value: 5.151e-09
```

With the non-significant p-value of the interaction term LogDurations:BeeWorker, there seems to be no conclusive evidence that the proportion of pollen depends on the duration of visit differently between queens and workers.

part e

```r
model_no_interact <- lm(Logit ~ LogDuration + Bee, data = ex0327)
summary(model_no_interact)
```

```
##
## Call:
## lm(formula = Logit ~ LogDuration + Bee, data = ex0327)
##
## Residuals:
##      Min       1Q   Median       3Q      Max
## -1.40852 -0.49627  0.08815  0.43598  1.15562
##
## Coefficients:
##             Estimate Std. Error t value Pr(>|t|)
## (Intercept)  -2.7146     0.3842  -7.065 9.18e-09 ***
## LogDuration   0.8886     0.1402   6.339 1.07e-07 ***
## BeeWorker     0.5697     0.2364   2.409   0.0202 *
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
##
## Residual standard error: 0.652 on 44 degrees of freedom
## Multiple R-squared:  0.6068, Adjusted R-squared:  0.5889
## F-statistic: 33.95 on 2 and 44 DF,  p-value: 1.206e-09
```

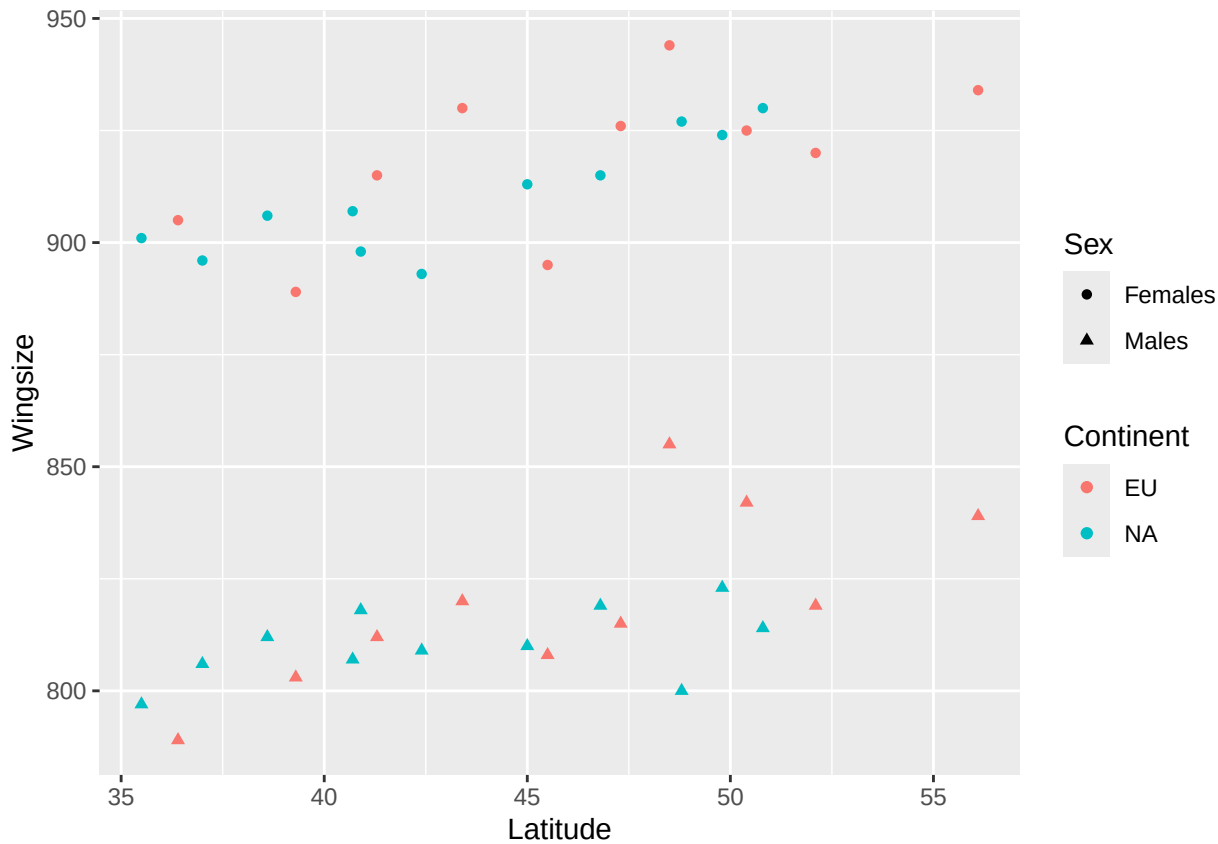
The BeeWorker coefficient is significant in model_no_interact, which means after adjusting for duration, the worker bees remove more pollen on average than queens.

This changed from the model_interact, from part d, because the interaction term allowed the queens and workers to have different slopes and intercepts. But in part e, we assumed workers and queens have the same slope, with different intercepts, which adjusts for duration.

Question 18

part a

```r
library(tidyr)

fly_long <- ex0918 |>
  pivot_longer(cols = c(Females, Males),
               names_to = "Sex",
               values_to = "Wingsize")

ggplot(fly_long, aes(x = Latitude, y = Wingsize, color = Continent, shape = Sex)) +
  geom_point()
```



part b

```
fly_model <- lm(Wingsize ~ Latitude * Continent + Sex, data = fly_long)
summary(fly_model)
```

```
##
## Call:
## lm(formula = Wingsize ~ Latitude * Continent + Sex, data = fly_long)
##
## Residuals:
##      Min       1Q   Median       3Q      Max
## -22.5537  -6.9902   0.7025   5.6303  29.9364
##
## Coefficients:
##                      Estimate Std. Error t value Pr(>|t|)
## (Intercept)          820.9865    19.8740  41.310  < 2e-16 ***
## Latitude               2.1224     0.4268   4.972 1.54e-05 ***
## ContinentNA           28.0224    28.3491   0.988    0.329
## SexMales             -98.8571     3.3977 -29.095  < 2e-16 ***
## Latitude:ContinentNA  -0.7217     0.6315  -1.143    0.260
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
##
## Residual standard error: 11.01 on 37 degrees of freedom
## Multiple R-squared:   0.96,  Adjusted R-squared:  0.9556
## F-statistic: 221.8 on 4 and 37 DF,  p-value: < 2.2e-16
```

Because the interaction term Latitude:ContinentNA is not significant, the comparison of wing size and latitude is the same across continents, so we can begin to conclude that the NA flies evolved the same latitude cline as the European flies.